

A Lower Bound for the Strong Arithmetic Regularity Lemma

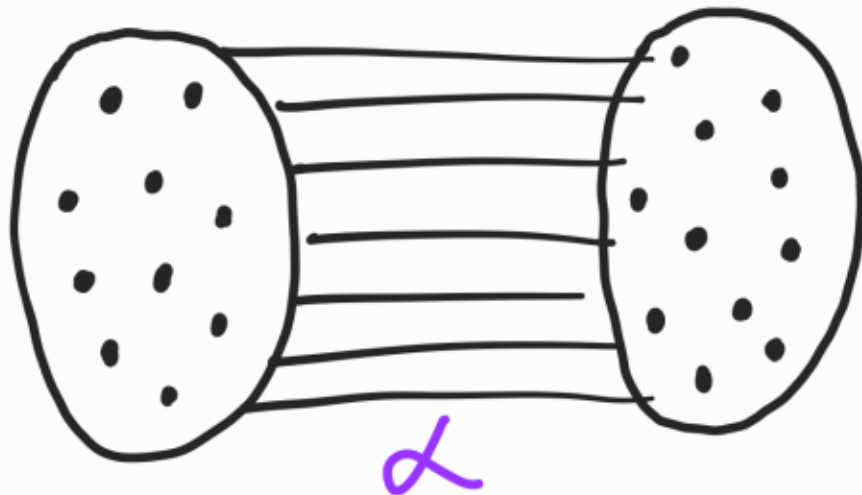
Szemerédi Regularity Lemma (Szemerédi, 1976)

For every $\varepsilon > 0$, any sufficiently large graph G has a vertex partition $V = V_1 \sqcup V_2 \sqcup \dots \sqcup V_{M(\varepsilon)}$ such that all but an ε -proportion of pairs (V_i, V_j) are ε -regular.

A Lower Bound for the Strong Arithmetic Regularity Lemma

Szemerédi Regularity Lemma (Szemerédi, 1976)

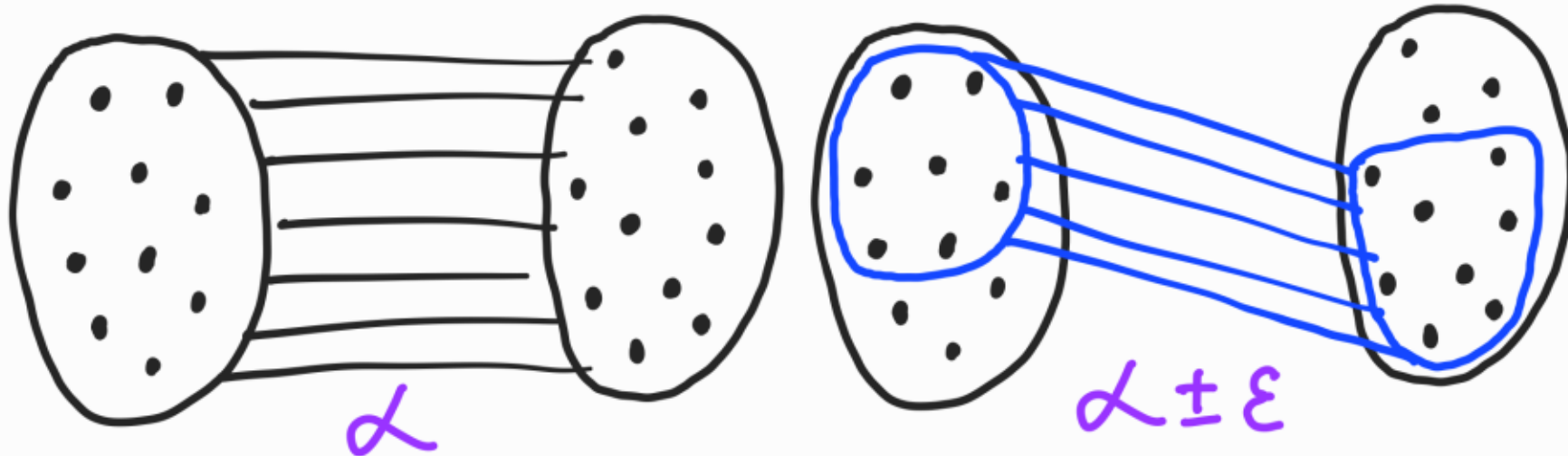
For every $\varepsilon > 0$, any sufficiently large graph G has a vertex partition $V = V_1 \sqcup V_2 \sqcup \dots \sqcup V_M(\varepsilon)$ such that all but an ε -proportion of pairs (V_i, V_j) are ε -regular.



A Lower Bound for the Strong Arithmetic Regularity Lemma

Szemerédi Regularity Lemma (Szemerédi, 1976)

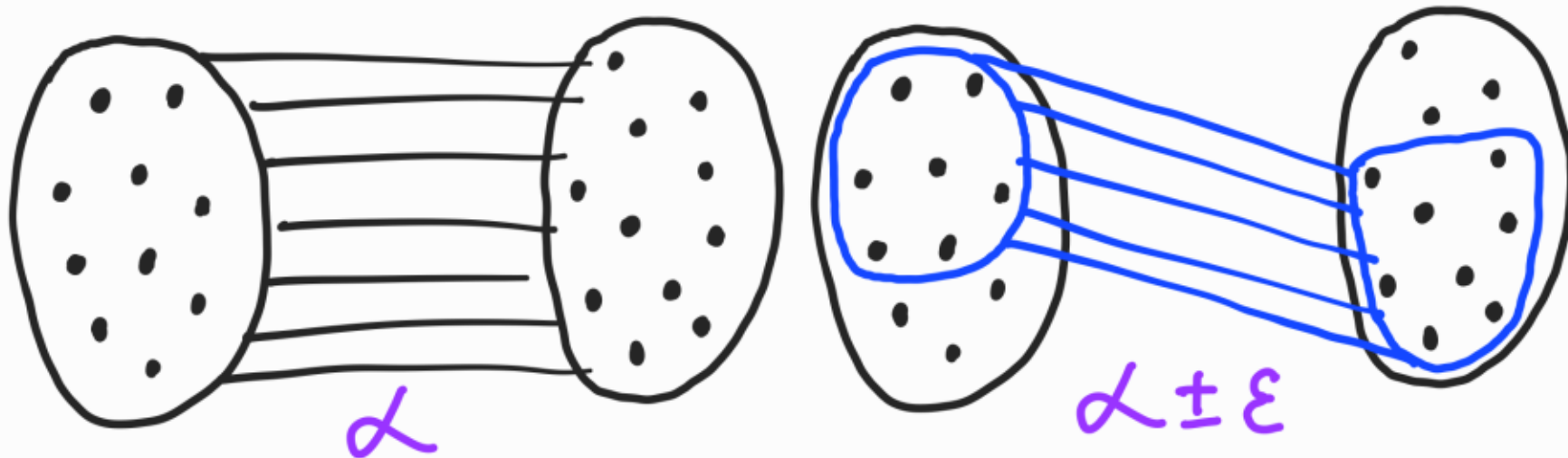
For every $\varepsilon > 0$, any sufficiently large graph G has a vertex partition $V = V_1 \sqcup V_2 \sqcup \dots \sqcup V_M(\varepsilon)$ such that all but an ε -proportion of pairs (V_i, V_j) are ε -regular.



A Lower Bound for the Strong Arithmetic Regularity Lemma

Szemerédi Regularity Lemma (Szemerédi, 1976)

For every $\varepsilon > 0$, any sufficiently large graph G has a vertex partition $V = V_1 \sqcup V_2 \sqcup \dots \sqcup V_{M(\varepsilon)}$ such that all but an ε -proportion of pairs (V_i, V_j) are ε -regular.



independent
of #vertices

A Lower Bound for the Strong Arithmetic Regularity Lemma

$V = V_1 \sqcup V_2 \sqcup \dots \sqcup V_M$ ← independent of #vertices

$$M = M(\varepsilon) \geq \text{twr}(\varepsilon^c) = 2^{2^{2^{\dots^2}}} \xrightarrow{\varepsilon^c} \varepsilon^{-c} \text{ (Gowers, 1997)}$$

A Lower Bound for the Strong Arithmetic Regularity Lemma

$V = V_1 \sqcup V_2 \sqcup \dots \sqcup V_M$ ← independent of #vertices

$$M = M(\varepsilon) \geq \text{twr}(\varepsilon^c) = 2^{2^{\dots^2}} \xrightarrow{\varepsilon^c} \varepsilon^{-c} \quad (\text{Gowers, 1997})$$

Application: triangle removal lemma
(Ruzsa, Szemerédi, 1976)

For all $\varepsilon > 0$, $\exists \delta > 0$ s.t. if a graph G has $\leq \delta n^3$ triangles, G can be made Δ -free by removing $\leq \varepsilon$ -proportion of edges.

A Lower Bound for the Strong Arithmetic Regularity Lemma

$V = V_1 \sqcup V_2 \sqcup \dots \sqcup V_M$ ← independent of #vertices

$$M = M(\varepsilon) \geq \text{twr}(\varepsilon^{-c}) = 2^{2^{\dots^2}} \xrightarrow{\varepsilon^{-c}} \varepsilon^{-c} \quad (\text{Gowers, 1997})$$

Application: triangle removal lemma
(Ruzsa, Szemerédi, 1976)

For all $\varepsilon > 0$, $\exists \delta > 0$ s.t. if a graph G has $\leq \delta n^3$ triangles, G can be made Δ -free by removing $\leq \varepsilon$ -proportion of edges.

$$\delta^{-1} \leq \text{twr}(\log(\varepsilon^{-1})) - \text{Fox, 2011}$$

A Lower Bound for the Strong Arithmetic Regularity Lemma

Induced subgraph removal lemma

(Alon, Fischer, Krivelevich, Szegedy, 2000)

For all $\varepsilon > 0$, $\exists \delta > 0$ s.t. if a graph G has $\leq \delta n^k$ induced copies of subgraph H on k vertices, G can be made H -free by modifying $\leq \varepsilon$ -proportion of edges.

A Lower Bound for the Strong Arithmetic Regularity Lemma

Strong Regularity Lemma

(Alon, Fischer, Krivelevich, Szegedy, 2000)

For every $\varepsilon: \mathbb{N} \rightarrow (0,1)$, any sufficiently large G has vertex partitions $\mathcal{B} \preceq \mathcal{A}$, with $\mathcal{A} = \{V_1, \dots, V_{m(\varepsilon)}\}$ and $\mathcal{B} = \bigcup_{i=1}^m \{V_{i,j} : j=1, \dots, \ell(\varepsilon)\}$ such that:

A Lower Bound for the Strong Arithmetic Regularity Lemma

Strong Regularity Lemma

(Alon, Fischer, Krivelevich, Szegedy, 2000)

For every $\varepsilon: \mathbb{N} \rightarrow (0,1)$, any sufficiently large G has vertex partitions $B \preceq \mathcal{A}$, with $\mathcal{A} = \{V_1, \dots, V_{m(\varepsilon)}\}$

and $B = \bigcup_{i=1}^m \{V_{i,j} : j=1, \dots, \ell(\varepsilon)\}$ such that:

- a) all but an $\varepsilon(m)$ -proportion of pairs $(V_{i,j}, V_{i',j'})$ are $\varepsilon(m)$ -regular;

A Lower Bound for the Strong Arithmetic Regularity Lemma

Strong Regularity Lemma

(Alon, Fischer, Krivelevich, Szegedy, 2000)

For every $\varepsilon: \mathbb{N} \rightarrow (0, 1)$, any sufficiently large G has vertex partitions $B \preceq A$, with $A = \{V_1, \dots, V_{m(\varepsilon)}\}$

and $B = \bigcup_{i=1}^m \{V_{i,j} : j=1, \dots, \ell(\varepsilon)\}$ such that:

a) all but an $\varepsilon(m)$ -proportion of pairs

$(V_{i,j}, V_{i',j'})$ are $\varepsilon(m)$ -regular;

b) for all but $\varepsilon(0)$ -proportion of $(i, i') \in [m]^2$,
all but $\varepsilon(0)$ -proportion of $(j, j') \in [\ell]^2$ has

$$|d(V_i, V_{i'}) - d(V_{i,j}, V_{i',j'})| \leq \varepsilon(0).$$

A Lower Bound for the Strong Arithmetic Regularity Lemma

Strong Regularity Lemma

(Alon, Fischer, Krivelevich, Szegedy, 2000)

For every $\varepsilon: \mathbb{N} \rightarrow (0,1)$, any sufficiently large G has vertex partitions $B \preceq \mathcal{A}$, with $\mathcal{A} = \{V_1, \dots, V_{m(\varepsilon)}\}$

and $B = \bigcup_{i=1}^m \{V_{i,j} : j=1, \dots, \ell(\varepsilon)\}$ such that:

a) all but an $\varepsilon(m)$ -proportion of pairs $(V_{i,j}, V_{i',j'})$ are $\varepsilon(m)$ -regular;

b) for all but $\varepsilon(0)$ -proportion of $(i, i') \in [m]^2$,
all but $\varepsilon(0)$ -proportion of $(j, j') \in [\ell]^2$ has

$$|d(V_i, V_{i'}) - d(V_{i,j}, V_{i',j'})| \leq \varepsilon(0).$$

$\nwarrow$ constant

} densities
are close
most of the
time

A Lower Bound for the Strong Arithmetic Regularity Lemma

Strong Regularity Lemma

(Alon, Fischer, Krivelevich, Szegedy, 2000)

For every $\varepsilon: \mathbb{N} \rightarrow (0,1)$, any sufficiently large G has vertex partitions $B \preceq A$, with $A = \{V_1, \dots, V_{m(\varepsilon)}\}$ and $B = \bigcup_{i=1}^m \{V_{i,j} : j=1, \dots, \ell(\varepsilon)\}$ such that: independent of #vertices

a) all but an $\varepsilon(m)$ -proportion of pairs $(V_{i,j}, V_{i',j'})$ are $\varepsilon(m)$ -regular;

b) for all but $\varepsilon(0)$ -proportion of $(i, i') \in [m]^2$, all but $\varepsilon(0)$ -proportion of $(j, j') \in [\ell]^2$ has $|d(V_i, V_{i'}) - d(V_{i,j}, V_{i',j'})| \leq \varepsilon(0)$.

← constant

} densities are close most of the time

A Lower Bound for the Strong Arithmetic Regularity Lemma

$$V = \underbrace{V_{1,1} \sqcup \dots \sqcup V_{1,\ell}} \sqcup \underbrace{V_{2,1} \sqcup \dots \sqcup V_{2,\ell}} \sqcup \dots \sqcup \underbrace{V_{m,1} \sqcup \dots \sqcup V_{m,\ell}}$$

$$M(\varepsilon) = m(\varepsilon) \ell(\varepsilon) \geq \text{WWZ}(F(\varepsilon)) \quad (\text{Conlon, Fox 2011})$$

$$\text{WWZ}(n) = \underbrace{\text{twr}(\text{twr}(\dots \text{twr}(1)))}_{n \text{ times}}$$

A Lower Bound for the Strong Arithmetic Regularity Lemma

$$V = \underbrace{V_{1,1} \sqcup \dots \sqcup V_{1,\ell}} \sqcup \underbrace{V_{2,1} \sqcup \dots \sqcup V_{2,\ell}} \sqcup \dots \sqcup \underbrace{V_{m,1} \sqcup \dots \sqcup V_{m,\ell}}$$

$$M(\varepsilon) = m(\varepsilon) \ell(\varepsilon) \geq \text{WWZ}(F(\varepsilon)) \quad (\text{Conlon, Fox 2011})$$

$$\text{WWZ}(n) = \underbrace{\text{twr}(\text{twr}(\dots \text{twr}(1)))}_{n \text{ times}}$$

Application: induced subgraph removal lemma
(Alon, Fischer, Krivelevich, Szegedy, 2000)

For all $\varepsilon > 0$, $\exists \delta > 0$ s.t. if a graph G has $\leq \delta n^k$ induced copies of subgraph H on k vertices, G can be made H -free by modifying $\leq \varepsilon$ -proportion of edges.

A Lower Bound for the Strong Arithmetic Regularity Lemma

$$V = \underbrace{V_{1,1} \sqcup \dots \sqcup V_{1,\ell}} \sqcup \underbrace{V_{2,1} \sqcup \dots \sqcup V_{2,\ell}} \sqcup \dots \sqcup \underbrace{V_{m,1} \sqcup \dots \sqcup V_{m,\ell}}$$

$$M(\varepsilon) = m(\varepsilon) \ell(\varepsilon) \geq \text{WWZ}(F(\varepsilon)) \quad (\text{Conlon, Fox 2011})$$

$$\text{WWZ}(n) = \underbrace{\text{twr}(\text{twr}(\dots \text{twr}(1)))}_{n \text{ times}}$$

Application: induced subgraph removal lemma
(Alon, Fischer, Krivelevich, Szegedy, 2000)

For all $\varepsilon > 0$, $\exists \delta > 0$ s.t. if a graph G has $\leq \delta n^k$ induced copies of subgraph H on k vertices, G can be made H -free by modifying $\leq \varepsilon$ -proportion of edges. tower-type bounds — Conlon, Fox

A Lower Bound for the Strong Arithmetic Regularity Lemma

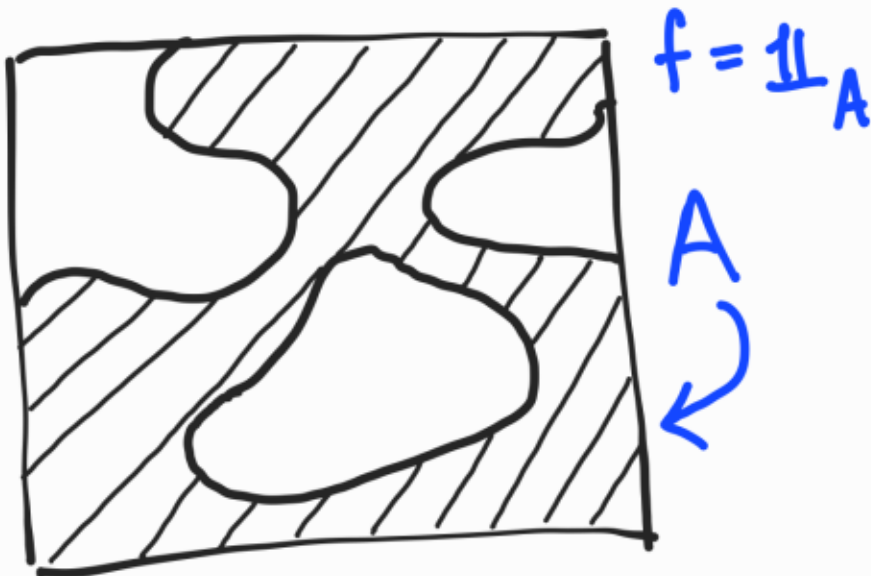
Arithmetic Regularity Lemma (Green, 2005)

For every $\varepsilon > 0$ and $f: \mathbb{F}_2^n \rightarrow [0, 1]$, there is a subspace $H \leq \mathbb{F}_2^n$ of bounded codimension such that f is ε -regular on all but an ε -proportion cosets of H .

A Lower Bound for the Strong Arithmetic Regularity Lemma

Arithmetic Regularity Lemma (Green, 2005)

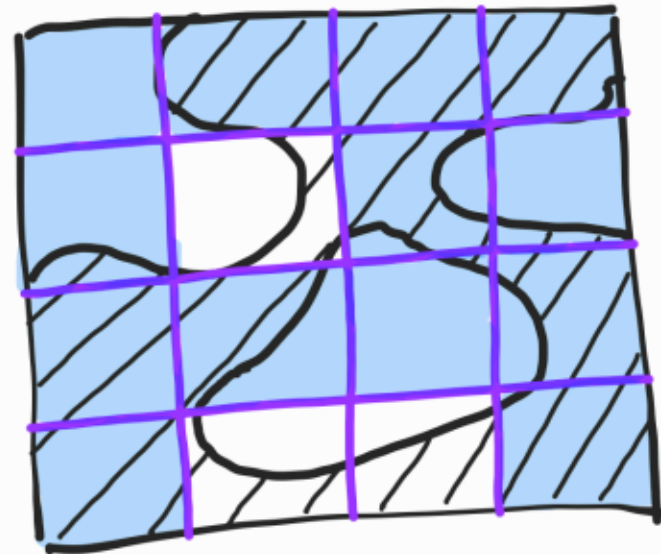
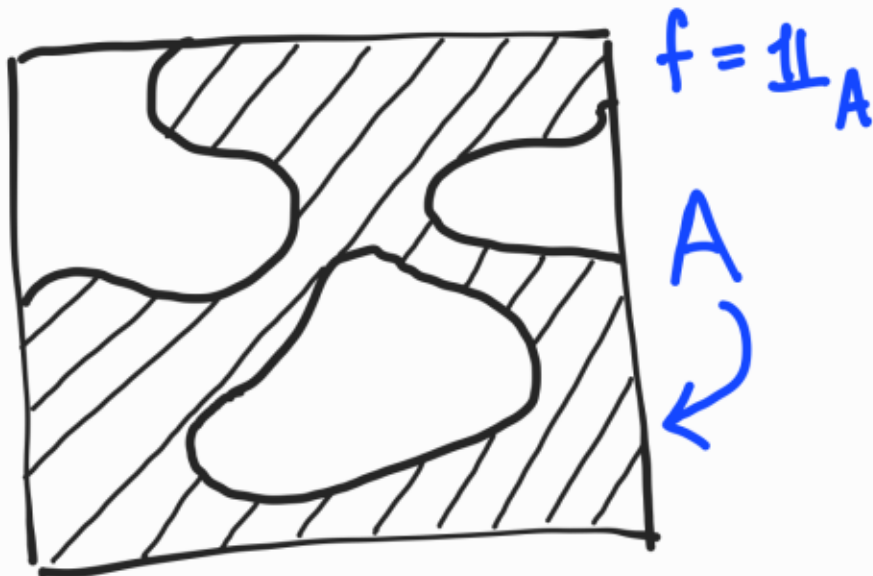
For every $\varepsilon > 0$ and $f: \mathbb{F}_2^n \rightarrow [0, 1]$, there is a subspace $H \leq \mathbb{F}_2^n$ of bounded codimension such that f is ε -regular on all but an ε -proportion cosets of H .



A Lower Bound for the Strong Arithmetic Regularity Lemma

Arithmetic Regularity Lemma (Green, 2005)

For every $\varepsilon > 0$ and $f: \mathbb{F}_2^n \rightarrow [0, 1]$, there is a subspace $H \leq \mathbb{F}_2^n$ of bounded codimension such that f is ε -regular on all but an ε -proportion cosets of H .

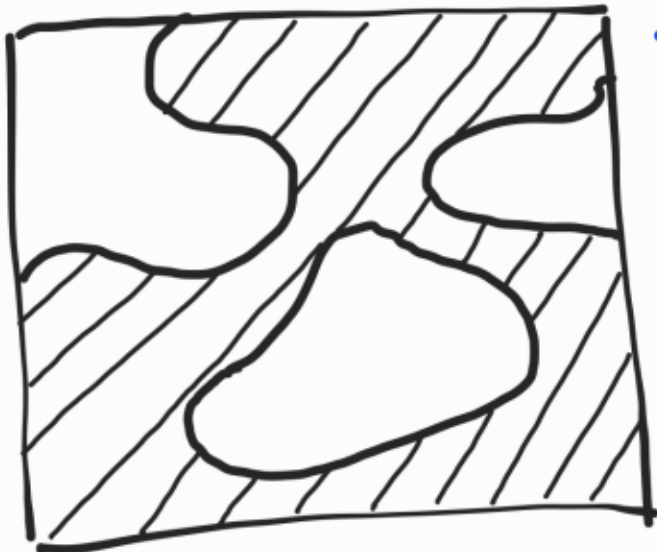


A Lower Bound for the Strong Arithmetic Regularity Lemma

Arithmetic Regularity Lemma (Green, 2005)

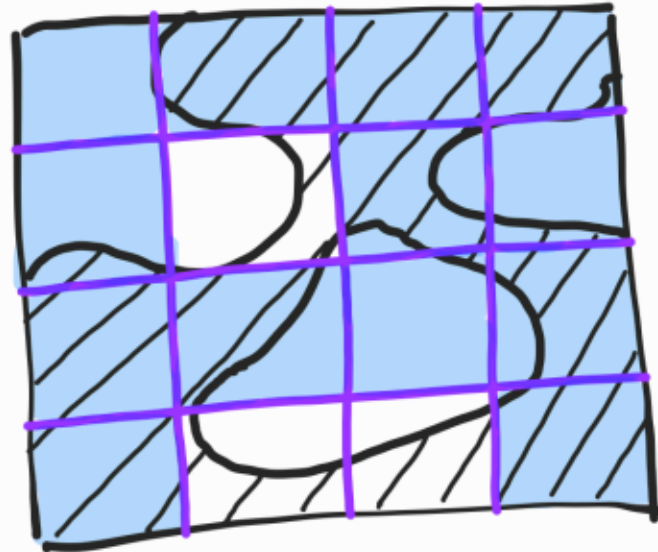
For every $\varepsilon > 0$ and $f: \mathbb{F}_2^n \rightarrow [0, 1]$, there is a subspace $H \leq \mathbb{F}_2^n$ of bounded codimension such that f is ε -regular on all but an ε -proportion cosets of H .

independent of n



$$f = \mathbb{1}_A$$

A

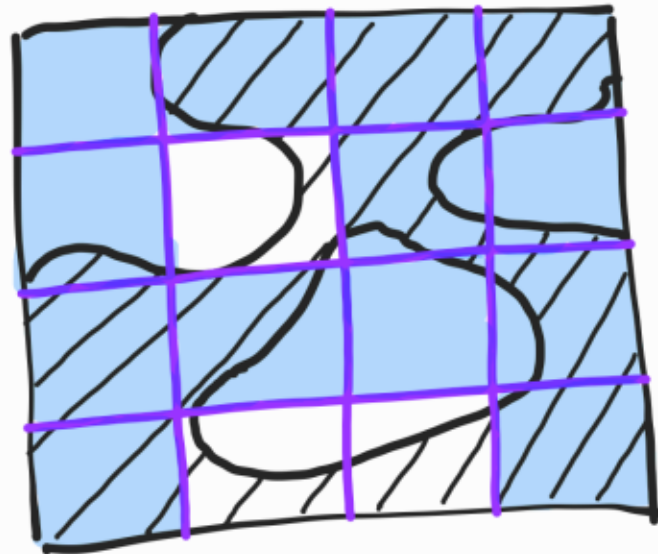
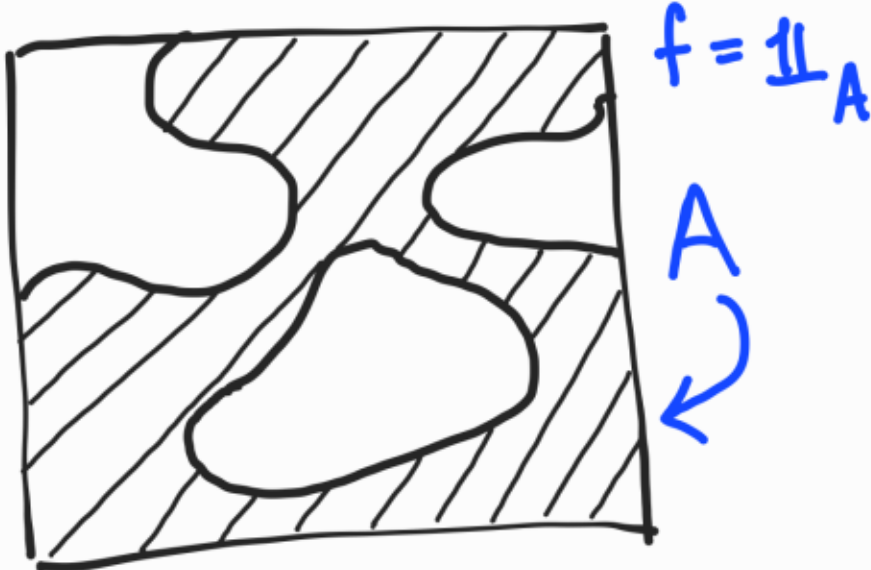


A Lower Bound for the Strong Arithmetic Regularity Lemma

Arithmetic Regularity Lemma (Green, 2005)

For every $\varepsilon > 0$ and $f: \mathbb{F}_2^n \rightarrow [0, 1]$, there is a subspace $H \leq \mathbb{F}_2^n$ of bounded codimension such that f is ε -regular on all but an ε -proportion cosets of H .

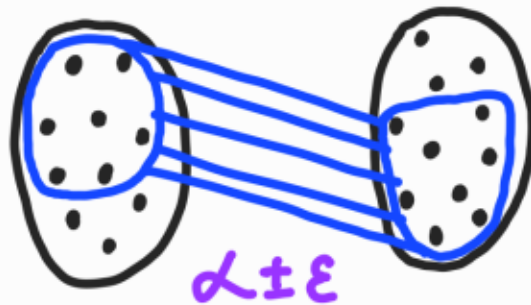
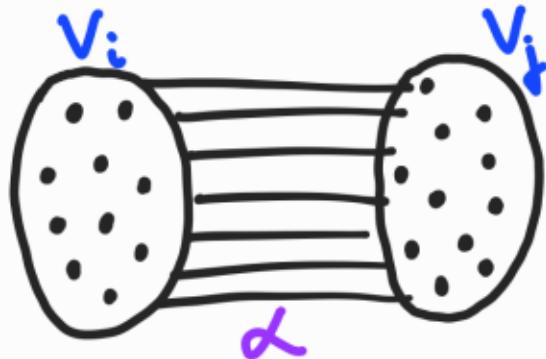
independent of n



A Lower Bound for the Strong Arithmetic Regularity Lemma

ε -regularity ($\mathbb{F}_2^n$)

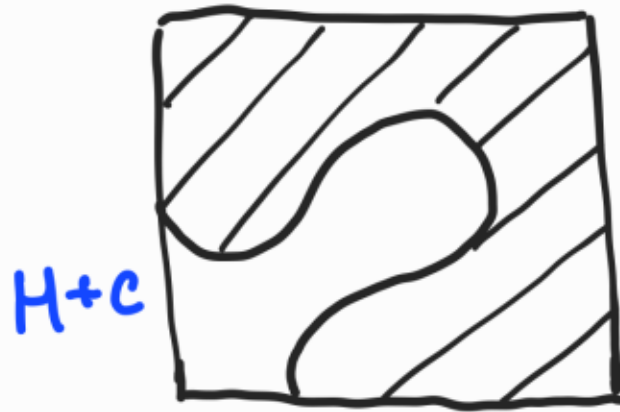
ε -regularity (graphs)



"no pair of large subsets has much higher/lower density"

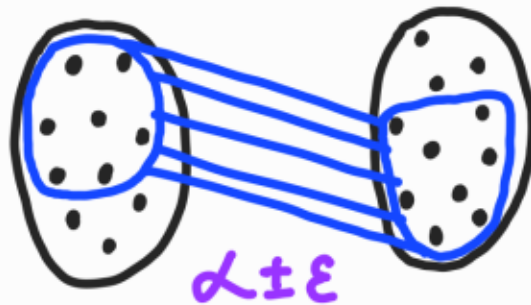
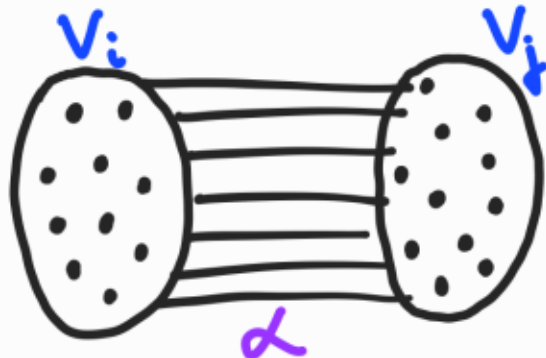
A Lower Bound for the Strong Arithmetic Regularity Lemma

ε -regularity ($\mathbb{F}_2^n$)



$$|\hat{f}_{H+C}^\wedge| \leq \varepsilon$$

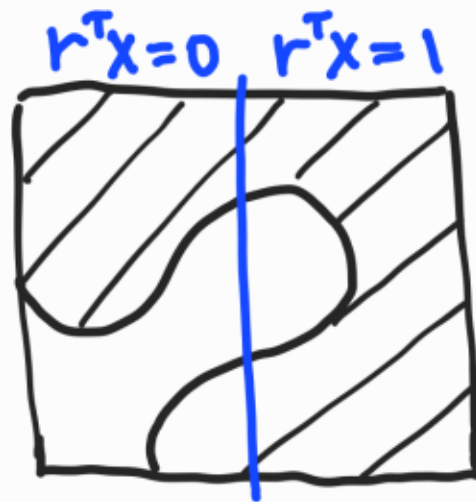
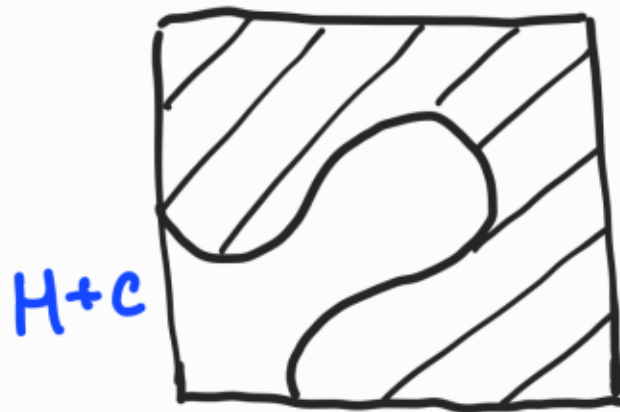
ε -regularity (graphs)



"no pair of large subsets has much higher/lower density"

A Lower Bound for the Strong Arithmetic Regularity Lemma

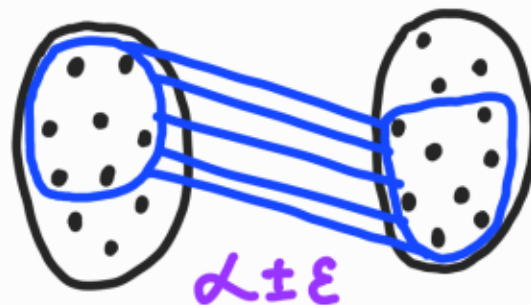
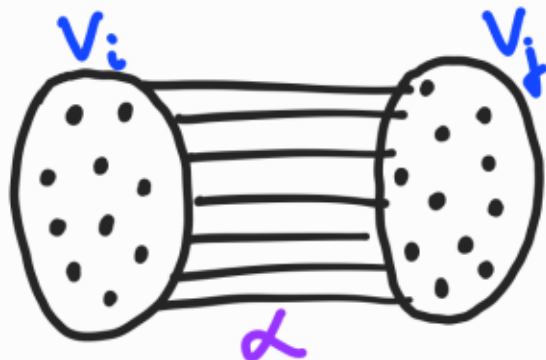
ε -regularity ($\mathbb{F}_2^n$)



$$|\hat{f}_{H+c}(r)| \leq \varepsilon$$

$$\hat{f}_{H+c}(r) = \mathbb{E}_{x \in H+c} f(x) (-1)^{r^T x}$$

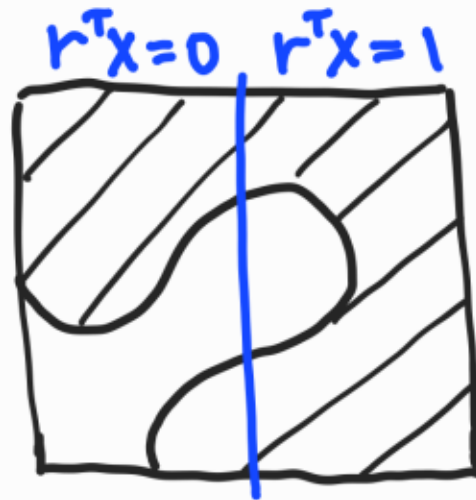
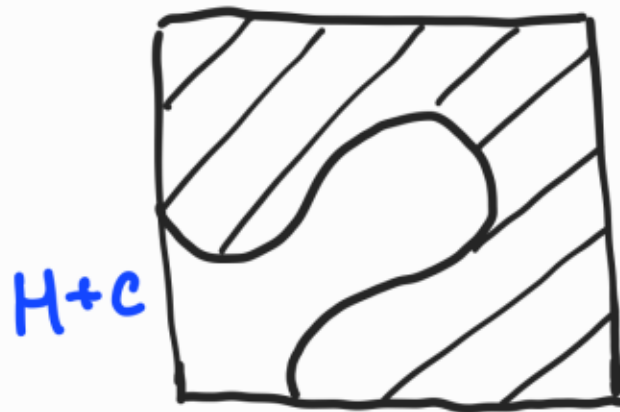
ε -regularity (graphs)



"no pair of large subsets has much higher/lower density"

A Lower Bound for the Strong Arithmetic Regularity Lemma

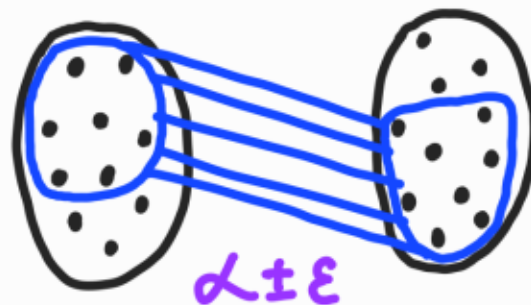
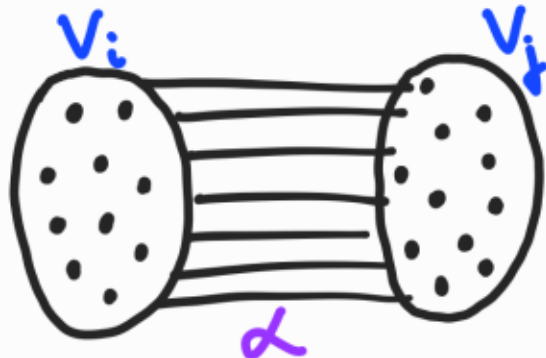
ε -regularity ($\mathbb{F}_2^n$)



$$|\hat{f}_{H+C}(r)| \leq \varepsilon$$

$$\begin{aligned} \hat{f}_{H+C}(r) &= \mathbb{E}_{x \in H+C} f(x) (-1)^{r^T x} \\ &= \mathbb{E} f(x) \mathbb{1}_{\varepsilon r^\perp}(x) - \mathbb{E} f(x) \mathbb{1}_{\varepsilon r^\perp}(x) \end{aligned}$$

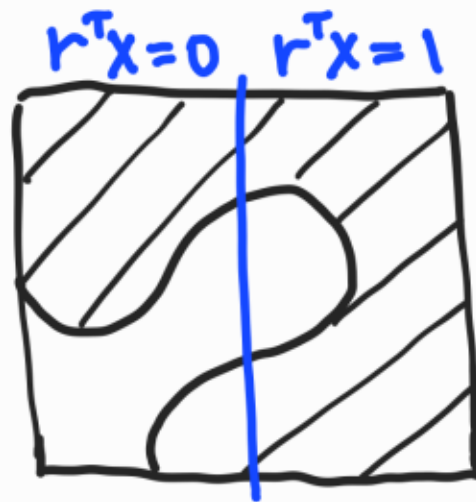
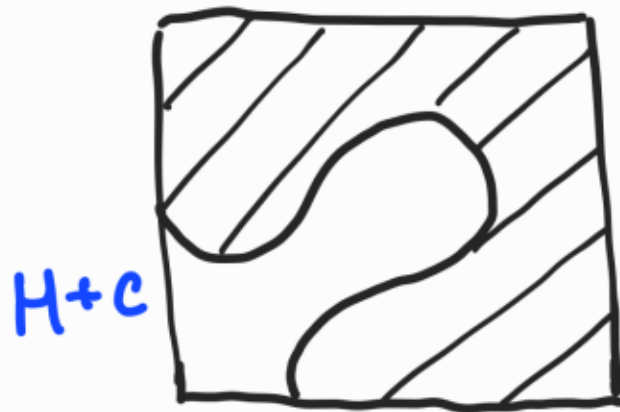
ε -regularity (graphs)



"no pair of large subsets has much higher/lower density"

A Lower Bound for the Strong Arithmetic Regularity Lemma

ε -regularity ($\mathbb{F}_2^n$)

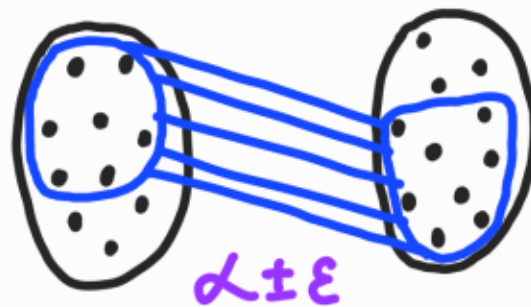
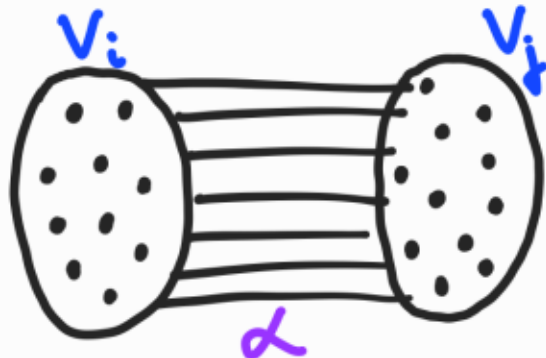


$$|\hat{f}_{H+C}(r)| \leq \varepsilon$$

$$\begin{aligned} \hat{f}_{H+C}(r) &= \mathbb{E}_{x \in H+C} f(x) (-1)^{r^T x} \\ &= \mathbb{E} f(x) \mathbb{1}_{\varepsilon r^\perp}(x) - \mathbb{E} f(x) \mathbb{1}_{\varepsilon r^\perp}(x) \end{aligned}$$

"no codim-1 subspace has much higher density"

ε -regularity (graphs)



"no pair of large subsets has much higher/lower density"

A Lower Bound for the Strong Arithmetic Regularity Lemma

Strong ARL (Bhattacharyya, Fischer, Lovett, 2013)

For every $\varepsilon: \mathbb{N} \rightarrow (0, 1)$ and $f: \mathbb{F}_2^n \rightarrow [0, 1]$, there are subspaces $H_2 \leq H_1 \leq \mathbb{F}_2^n$ of bounded codimensions $D_2 \geq D_1$ such that:

i) f is $\varepsilon(D_1)$ -regular on all but an $\varepsilon(D_1)$ -proportion of cosets of H_2 ;

ii) $\mathcal{E}(H_2) - \mathcal{E}(H_1) \leq \varepsilon(0)$.

" H_2 -cosets mostly agree in density with H_1 -cosets"

A Lower Bound for the Strong Arithmetic Regularity Lemma

Strong ARL (Bhattacharyya, Fischer, Lovett, 2013)

For every $\varepsilon: \mathbb{N} \rightarrow (0, 1)$ and $f: \mathbb{F}_2^n \rightarrow [0, 1]$, there are subspaces $H_2 \leq H_1 \leq \mathbb{F}_2^n$ of bounded codimensions $D_2 \geq D_1$ such that:

i) f is $\varepsilon(D_1)$ -regular on all but an $\varepsilon(D_1)$ -proportion of cosets of H_2 ;

ii) $\mathcal{E}(H_2) - \mathcal{E}(H_1) \leq \varepsilon(0)$.

" H_2 -cosets mostly agree in density with H_1 -cosets"

energy

$$\mathcal{E}(H_2) = \|\mathbb{E}(f|H_2)\|_{L_2}^2 = \mathbb{E}_x d_{H_2+c_x}^2$$

"average square density on cosets of H_2 "

A Lower Bound for the Strong Arithmetic Regularity Lemma

Strong ARL (Bhattacharyya, Fischer, Lovett, 2013)

For every $\varepsilon: \mathbb{N} \rightarrow (0, 1)$ and $f: \mathbb{F}_2^n \rightarrow [0, 1]$, there are subspaces $H_2 \leq H_1 \leq \mathbb{F}_2^n$ of bounded codimensions $D_2 \geq D_1$ such that:

i) f is $\varepsilon(D_1)$ -regular on all but an $\varepsilon(D_1)$ -proportion of cosets of H_2 ;

ii) $\mathcal{E}(H_2) - \mathcal{E}(H_1) \leq \varepsilon(0)$.

energy

$$\mathcal{E}(H_2) = \|\mathbb{E}(f|H_2)\|_{L_2}^2 = \mathbb{E}_x d_{H_2+cx}^2$$

"average square density on cosets of H_2 "

" H_2 -cosets mostly agree in density with H_1 -cosets"

Pythagoras' Theorem

$$\mathcal{E}(H_2) - \mathcal{E}(H_1) = \|\mathbb{E}(f|H_2) - \mathbb{E}(f|H_1)\|_{L_2}^2$$

"average square density difference"

A Lower Bound for the Strong Arithmetic Regularity Lemma

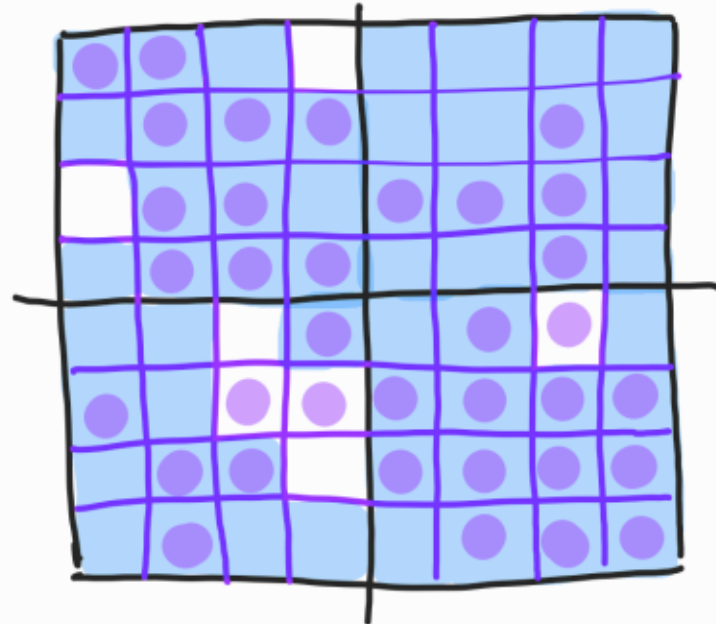
Strong ARL (Bhattacharyya, Fischer, Lovett, 2013)

For every $\varepsilon: \mathbb{N} \rightarrow (0, 1)$ and $f: \mathbb{F}_2^n \rightarrow [0, 1]$, there are subspaces $H_2 \leq H_1 \leq \mathbb{F}_2^n$ of bounded codimensions $D_2 \geq D_1$ such that:

i) f is $\varepsilon(D_1)$ -regular on all but an $\varepsilon(D_1)$ -proportion of cosets of H_2 ;

ii) $\mathcal{E}(H_2) - \mathcal{E}(H_1) \leq \varepsilon(0)$.

" H_2 -cosets mostly agree in density with H_1 -cosets"



A Lower Bound for the Strong Arithmetic Regularity Lemma

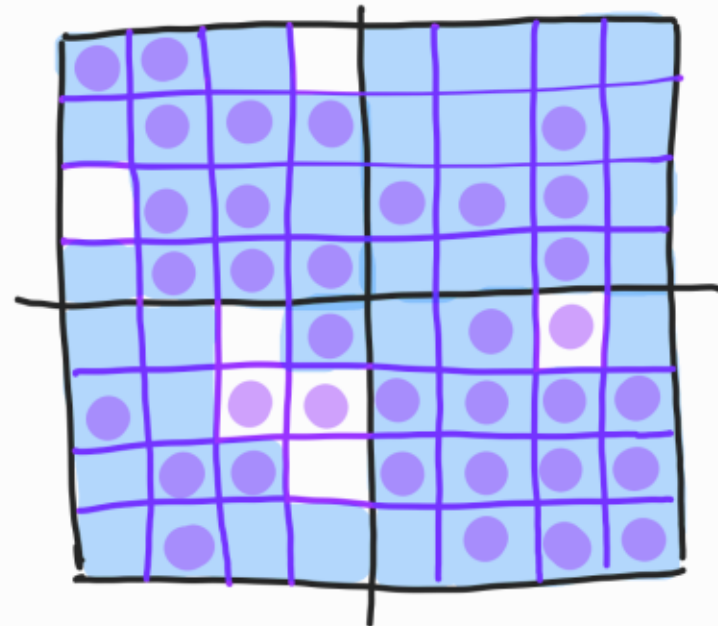
Strong ARL (Bhattacharyya, Fischer, Lovett, 2013)

For every $\varepsilon: \mathbb{N} \rightarrow (0, 1)$ and $f: \mathbb{F}_2^n \rightarrow [0, 1]$, there are subspaces $H_2 \leq H_1 \leq \mathbb{F}_2^n$ of bounded codimensions $D_2 \geq D_1$ such that:

i) f is $\varepsilon(D_1)$ -regular on all but an $\varepsilon(D_1)$ -proportion of cosets of H_2 ;

ii) $\mathcal{E}(H_2) - \mathcal{E}(H_1) \leq \varepsilon(0)$.

" H_2 -cosets mostly agree in density with H_1 -cosets"



} d

A Lower Bound for the Strong Arithmetic Regularity Lemma

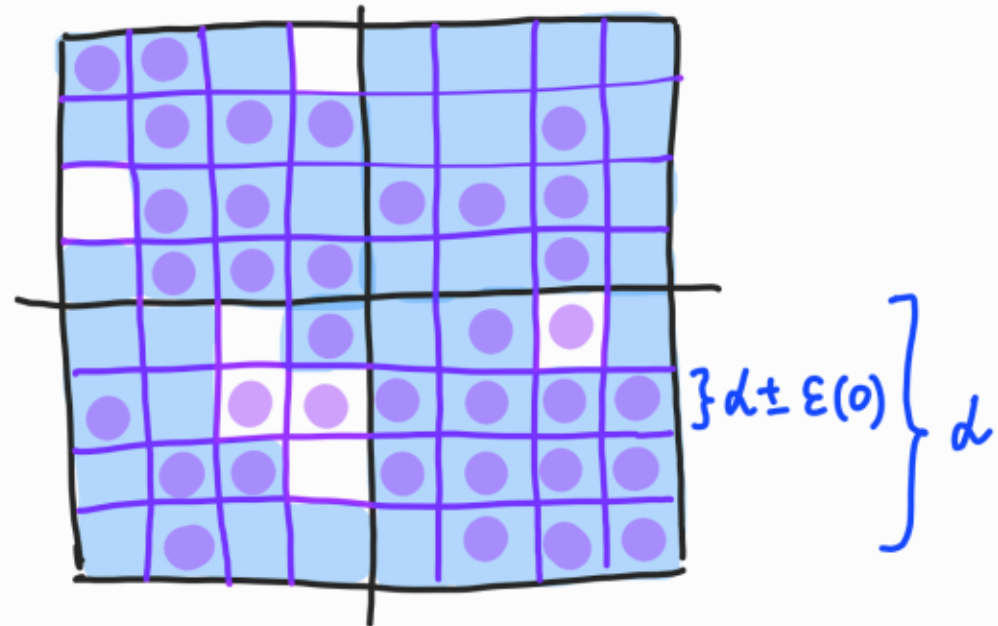
Strong ARL (Bhattacharyya, Fischer, Lovett, 2013)

For every $\varepsilon: \mathbb{N} \rightarrow (0, 1)$ and $f: \mathbb{F}_2^n \rightarrow [0, 1]$, there are subspaces $H_2 \leq H_1 \leq \mathbb{F}_2^n$ of bounded codimensions $D_2 \geq D_1$ such that:

i) f is $\varepsilon(D_1)$ -regular on all but an $\varepsilon(D_1)$ -proportion of cosets of H_2 ;

ii) $\mathcal{E}(H_2) - \mathcal{E}(H_1) \leq \varepsilon(0)$.

" H_2 -cosets mostly agree in density with H_1 -cosets"



A Lower Bound for the Strong Arithmetic Regularity Lemma

Applications:

- Arithmetic removal lemma (Green, 2005), (Shapira, 2010)
- Induced arithmetic removal lemma (Bhattacharyya, Fischer, Hatami $\times 2$, Lovett, 2013)
(Fox, Tidor, Zhao 2022), (G., 2024+)

A Lower Bound for the Strong Arithmetic Regularity Lemma

Applications:

- Arithmetic removal lemma (Green, 2005), (Shapira, 2010)
- Induced arithmetic removal lemma (Bhattacharyya, Fischer, Hatami $\times 2$, Lovett, 2013)
(Fox, Tidor, Zhao 2022), (G., 2024+)

Bounds:

	Graphs	Arithmetic
Regularity	$\text{twr}(\varepsilon^c)$	
Strong Regularity	$\text{wwz}(F(\varepsilon))$	

A Lower Bound for the Strong Arithmetic Regularity Lemma

Applications:

- Arithmetic removal lemma (Green, 2005), (Shapira, 2010)
- Induced arithmetic removal lemma (Bhattacharyya, Fischer, Hatami $\times 2$, Lovett, 2013)
(Fox, Tidor, Zhao 2022), (G., 2024+)

Bounds:

	Graphs	Arithmetic
Regularity	$\text{twr}(\varepsilon^{-c})$	$\text{twr}(\varepsilon^{-1})$ (Green, 2005) (Hosseini, Lovett, Moshkovitz, Shapira, 2014)
Strong Regularity	$\text{wwz}(F(\varepsilon))$	

A Lower Bound for the Strong Arithmetic Regularity Lemma

Applications:

- Arithmetic removal lemma (Green, 2005), (Shapira, 2010)
- Induced arithmetic removal lemma (Bhattacharyya, Fischer, Hatami $\times 2$, Lovett, 2013)
(Fox, Tidor, Zhao 2022), (G., 2024+)

Bounds:

	Graphs	Arithmetic
Regularity	$\text{twr}(\varepsilon^{-c})$	$\text{twr}(\varepsilon^{-1})$ (Green, 2005) (Hosseini, Lovett, Moshkovitz, Shapira, 2014)
Strong Regularity	$\text{WWZ}(F(\varepsilon))$	$\text{WWZ}(F(\varepsilon))$ (G., 2024+)

A Lower Bound for the Strong Arithmetic Regularity Lemma

Applications:

- Arithmetic removal lemma (Green, 2005), (Shapira, 2010)
- Induced arithmetic removal lemma (Bhattacharyya, Fischer, Hatami $\times 2$, Lovett, 2013)
(Fox, Tidor, Zhao 2022), (G., 2024+)

Bounds:

	Graphs	Arithmetic
Regularity	$\text{twr}(\varepsilon^{-c})$	$\text{twr}(\varepsilon^{-1})$ (Green, 2005) (Hosseini, Lovett, Moshkovitz, Shapira, 2014)
Strong Regularity	$\text{wwz}(F(\varepsilon))$	$\text{wwz}(F(\varepsilon))$ (G., 2024+) $\exists f: \mathbb{F}_2^n \rightarrow [0,1]$ s.t. D_2 must be $\geq \text{wwz}$

A Lower Bound for the Strong Arithmetic Regularity Lemma

$$d_{i+1} = \begin{cases} 2^{D_i} & \text{if } i \leq 3; \\ \frac{1}{8} 2^{D_i} & \text{otherwise.} \end{cases}$$

A Lower Bound for the Strong Arithmetic Regularity Lemma

$$d_{i+1} = \begin{cases} 2^{D_i} & \text{if } i \leq 3; \\ \frac{1}{8} 2^{D_i} & \text{otherwise.} \end{cases}$$

$$D_0 = 0, \quad D_i = \sum_{j=1}^i d_j$$

A Lower Bound for the Strong Arithmetic Regularity Lemma

$$d_{i+1} = \begin{cases} 2^{D_i} & \text{if } i \leq 3; \\ \frac{1}{8} 2^{D_i} & \text{otherwise.} \end{cases}$$

$$D_0 = 0, \quad D_i = \sum_{j=1}^i d_j$$

$$d_i = 1, 2, 8, 2048, 2^{2056}, \dots$$

A Lower Bound for the Strong Arithmetic Regularity Lemma

$$d_{i+1} = \begin{cases} 2^{D_i} & \text{if } i \leq 3; \\ \frac{1}{8} 2^{D_i} & \text{otherwise.} \end{cases}$$

$$D_0 = 0, \quad D_i = \sum_{j=1}^i d_j$$

$$d_i = 1, 2, 8, 2048, 2^{2056}, \dots$$

$$d_i \geq \text{twr}(i-1)$$

A Lower Bound for the Strong Arithmetic Regularity Lemma

$$d_{i+1} = \begin{cases} 2^{D_i} & \text{if } i \leq 3; \\ \frac{1}{8} 2^{D_i} & \text{otherwise.} \end{cases}$$

$$d_i = 1, 2, 8, 2048, 2^{2056}, \dots$$

$$d_i \geq \text{twr}(i-1)$$

$$D_0 = 0, \quad D_i = \sum_{j=1}^i d_j$$

Lemma (Hosseini, Lovett, Moshkovitz, Shapira, 2014)

There is a set of $8d$ vectors in $\mathbb{F}_2^d$ such that any $\frac{3}{4}$ -proportion of them spans $\mathbb{F}_2^d$.

A Lower Bound for the Strong Arithmetic Regularity Lemma

Lemma

There are $8d$ vectors in $\mathbb{F}_2^d$
s.t. any $\frac{3}{4}$ -proportion spans $\mathbb{F}_2^d$.

$$d_{i+1} = \begin{cases} 2^{D_i} & \text{if } i \leq 3; \\ \frac{1}{8} 2^{D_i} & \text{otherwise.} \end{cases}$$
$$D_0 = 0, D_i = \sum_{j=1}^i d_j$$

A Lower Bound for the Strong Arithmetic Regularity Lemma

Lemma

There are $8d$ vectors in $\mathbb{F}_2^d$
s.t. any $\frac{3}{4}$ -proportion spans $\mathbb{F}_2^d$.

$$\mathbb{F}_2^n = H_0 \supseteq H_1 \supseteq \dots \supseteq H_s$$

$$\text{codim}_{H_i} H_{i+1} = d_{i+1}$$

$$d_{i+1} = \begin{cases} 2^{D_i} & \text{if } i \leq 3; \\ \frac{1}{8} 2^{D_i} & \text{otherwise.} \end{cases}$$
$$D_0 = 0, D_i = \sum_{j=1}^i d_j$$

A Lower Bound for the Strong Arithmetic Regularity Lemma

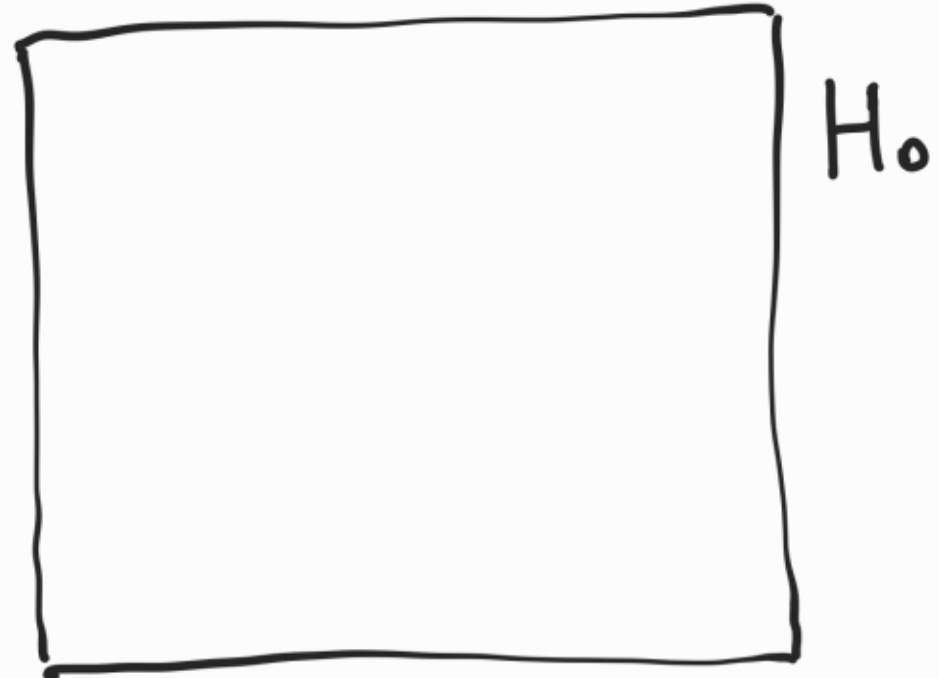
Lemma

There are $8d$ vectors in $\mathbb{F}_2^d$
s.t. any $\frac{3}{4}$ -proportion spans $\mathbb{F}_2^d$.

$$\mathbb{F}_2^n = H_0 \supseteq H_1 \supseteq \dots \supseteq H_s$$

$$\text{codim}_{H_i} H_{i+1} = d_{i+1}$$

$$d_{i+1} = \begin{cases} 2^{D_i} & \text{if } i \leq 3; \\ \frac{1}{8} 2^{D_i} & \text{otherwise.} \end{cases}$$
$$D_0 = 0, D_i = \sum_{j=1}^i d_j$$



A Lower Bound for the Strong Arithmetic Regularity Lemma

Lemma

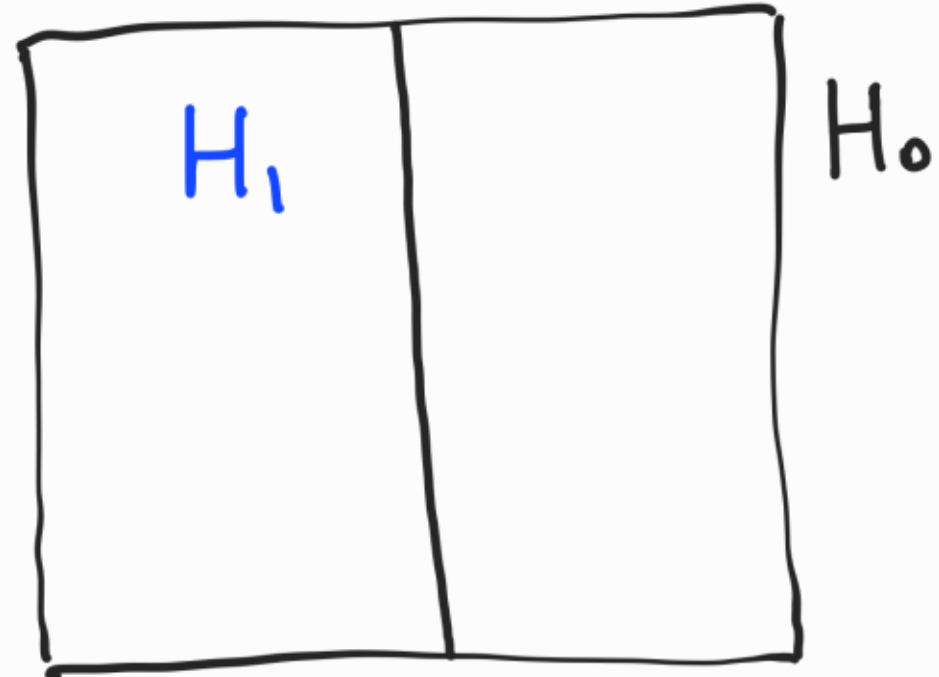
There are $8d$ vectors in $\mathbb{F}_2^d$
s.t. any $\frac{3}{4}$ -proportion spans $\mathbb{F}_2^d$.

$$\mathbb{F}_2^n = H_0 \geq H_1 \geq \dots \geq H_s$$

$$\text{codim}_{H_i} H_{i+1} = d_{i+1}$$

$\text{codim } H_1 = 1$, 2 cosets

$$d_{i+1} = \begin{cases} 2^{D_i} & \text{if } i \leq 3; \\ \frac{1}{8} 2^{D_i} & \text{otherwise.} \end{cases}$$
$$D_0 = 0, D_i = \sum_{j=1}^i d_j$$



A Lower Bound for the Strong Arithmetic Regularity Lemma

Lemma

There are $8d$ vectors in $\mathbb{F}_2^d$
s.t. any $\frac{3}{4}$ -proportion spans $\mathbb{F}_2^d$.

$$\mathbb{F}_2^n = H_0 \supseteq H_1 \supseteq \dots \supseteq H_s$$

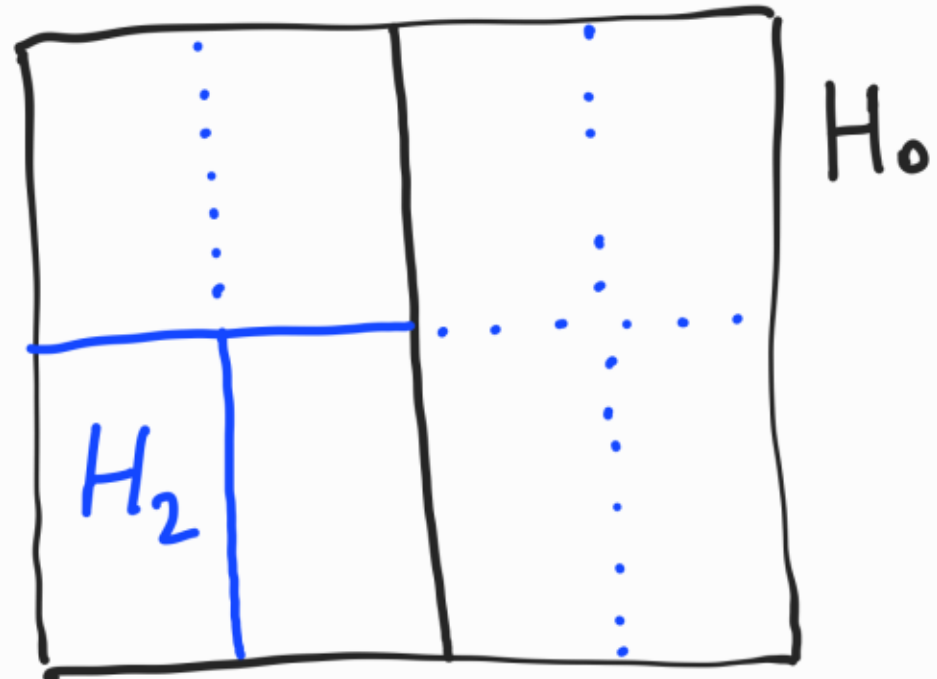
$$\text{codim}_{H_i} H_{i+1} = d_{i+1}$$

$\text{codim } H_1 = 1, 2 \text{ cosets}$

$\text{codim } H_2 = 1+2, 2^3 \text{ cosets}$

$$d_{i+1} = \begin{cases} 2^{D_i} & \text{if } i \leq 3; \\ \frac{1}{8} 2^{D_i} & \text{otherwise.} \end{cases}$$

$$D_0 = 0, D_i = \sum_{j=1}^i d_j$$



A Lower Bound for the Strong Arithmetic Regularity Lemma

Lemma.

Lemma

There are $8d$ vectors in $\mathbb{F}_2^d$
s.t. any $\frac{3}{4}$ -proportion spans $\mathbb{F}_2^d$.

$$H_2^n = H_0 \geq H_1 \geq \dots \geq H_s$$

$$\text{codim}_{H_i} H_{i+1} = d_{i+1}$$

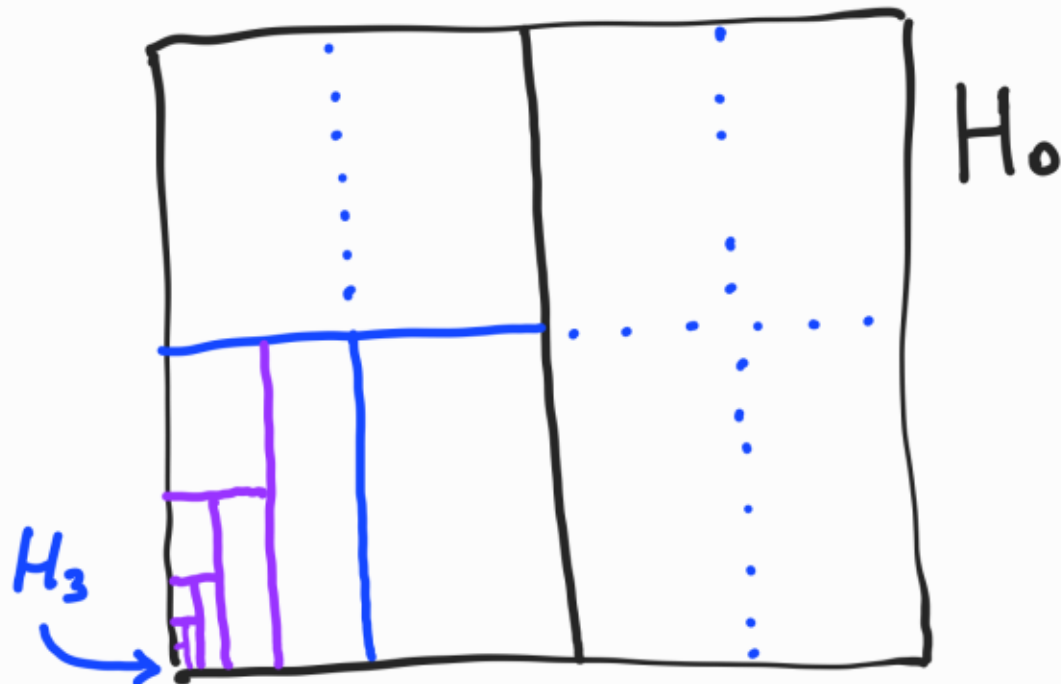
codim $H_1 = 1$, 2 cosets

$\text{codim } H_2 = 1+2, 2^3 \text{ cosets}$

$\text{codim } H_3 = 3 + 8, 2^{12} \text{ cosets}$

$$d_{i+1} = \begin{cases} 2^{D_i} & \text{if } i \leq 3; \\ \frac{1}{8} 2^{D_i} & \text{otherwise.} \end{cases}$$

$$D_0 = 0, \quad D_i = \sum_{j=1}^i d_j$$



A Lower Bound for the Strong Arithmetic Regularity Lemma

Lemma

There are $8d$ vectors in $\mathbb{F}_2^d$ s.t. any $\frac{3}{4}$ -proportion spans $\mathbb{F}_2^d$.

$$\mathbb{F}_2^n = H_0 \geq H_1 \geq \dots \geq H_s$$

$$\text{codim}_{H_i} H_{i+1} = d_{i+1}$$

$$\text{codim } H_1 = 1, 2 \text{ cosets}$$

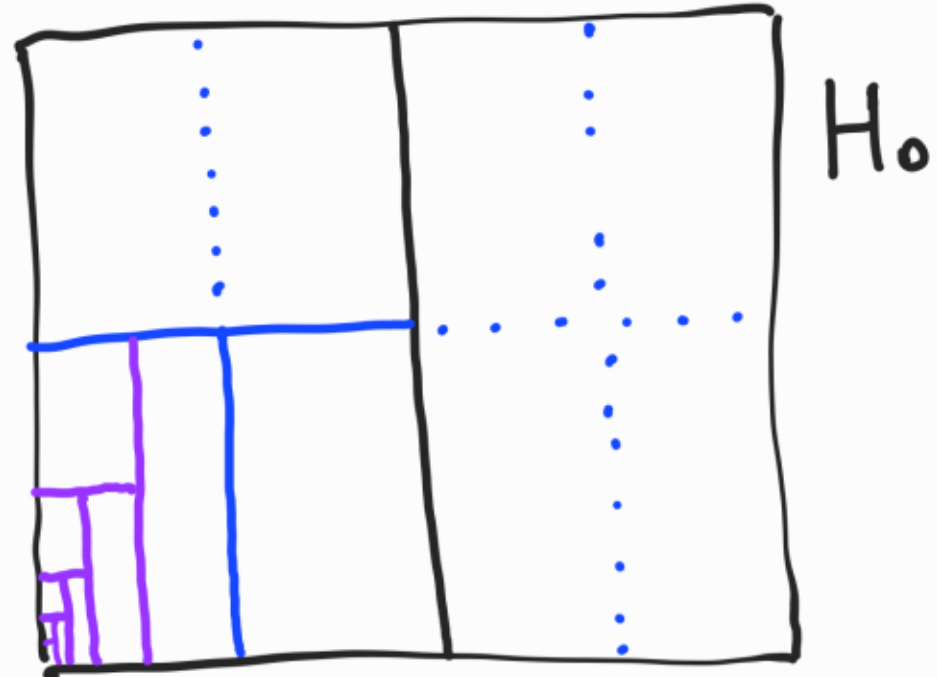
$$\text{codim } H_2 = 1+2, 2^3 \text{ cosets}$$

$$\text{codim } H_3 = 3+8, 2^{12} \text{ cosets}$$

$$\text{codim } H_i = D_i, 2^{D_i} = 8d_{i+1} \text{ cosets}$$

$$d_{i+1} = \begin{cases} 2^{D_i} & \text{if } i \leq 3; \\ \frac{1}{8} 2^{D_i} & \text{otherwise.} \end{cases}$$

$$D_0 = 0, D_i = \sum_{j=1}^i d_j$$



A Lower Bound for the Strong Arithmetic Regularity Lemma

$$\mathbb{F}_2^n = H_0 \supseteq H_1 \supseteq \dots \supseteq H_s$$

$\text{codim } H_i = D_i, 2^{D_i} = 8d_{i+1} \text{ cosets}$

Lemma

There are $8d$ vectors in $\mathbb{F}_2^d$
s.t. any $\frac{3}{4}$ -proportion spans $\mathbb{F}_2^d$.

A Lower Bound for the Strong Arithmetic Regularity Lemma

$$\mathbb{F}_2^n = H_0 \supseteq H_1 \supseteq \dots \supseteq H_s$$

$\text{codim } H_i = D_i, 2^{D_i} = 8d_{i+1}$ cosets

Pick U_i s.t. $H_i \oplus U_i = \mathbb{F}_2^n$

cosets of $H_i = \{H_i + u, u \in U_i\}, U_i \sim \mathbb{F}_2^{D_i}$

Lemma

There are $8d$ vectors in $\mathbb{F}_2^d$
s.t. any $\frac{3}{4}$ -proportion spans $\mathbb{F}_2^d$.

A Lower Bound for the Strong Arithmetic Regularity Lemma

$$\mathbb{F}_2^n = H_0 \supseteq H_1 \supseteq \dots \supseteq H_s$$

$\text{codim } H_i = D_i$, $2^{D_i} = 8d_{i+1}$ cosets

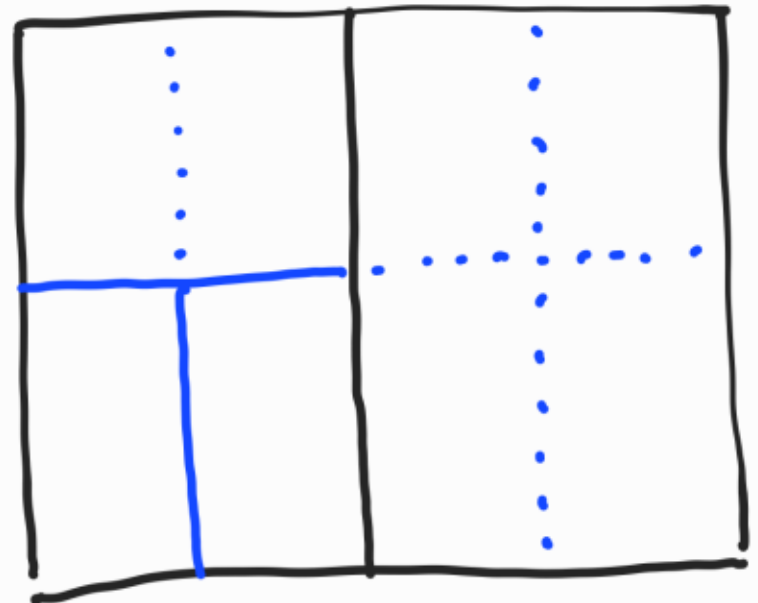
Pick U_i s.t. $H_i \oplus U_i = \mathbb{F}_2^n$

cosets of $H_i = \{H_i + u, u \in U_i\}$, $U_i \sim \mathbb{F}_2^{D_i}$

Pick codim-1 subspace
in each $H_i + u$

Lemma

There are $8d$ vectors in $\mathbb{F}_2^d$
s.t. any $\frac{3}{4}$ -proportion spans $\mathbb{F}_2^d$.



A Lower Bound for the Strong Arithmetic Regularity Lemma

$$\mathbb{F}_2^n = H_0 \supseteq H_1 \supseteq \dots \supseteq H_s$$

$\text{codim } H_i = D_i$, $2^{D_i} = 8d_{i+1}$ cosets

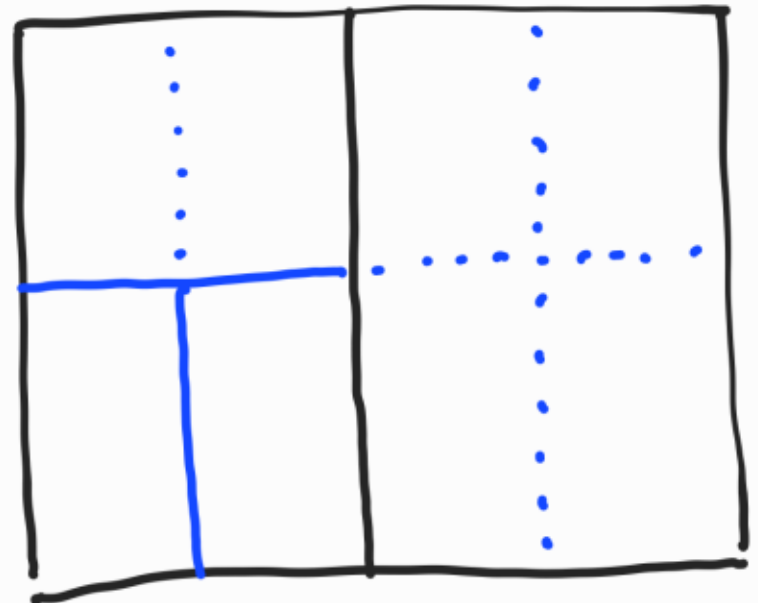
Pick U_i s.t. $H_i \oplus U_i = \mathbb{F}_p^n$

cosets of $H_i = \{H_i + u, u \in U_i\}$, $U_i \sim \mathbb{F}_p^{D_i}$

Pick codim-1 subspace
in each $H_i + u$: $r_{u,i}^T x = 0$

Lemma

There are $8d$ vectors in $\mathbb{F}_2^d$
s.t. any $\frac{3}{4}$ -proportion spans $\mathbb{F}_2^d$.



A Lower Bound for the Strong Arithmetic Regularity Lemma

$$\mathbb{F}_2^n = H_0 \supseteq H_1 \supseteq \dots \supseteq H_s$$

$\text{codim } H_i = D_i$, $2^{D_i} = 8d_{i+1}$ cosets

Pick U_i s.t. $H_i \oplus U_i = \mathbb{F}_p^n$

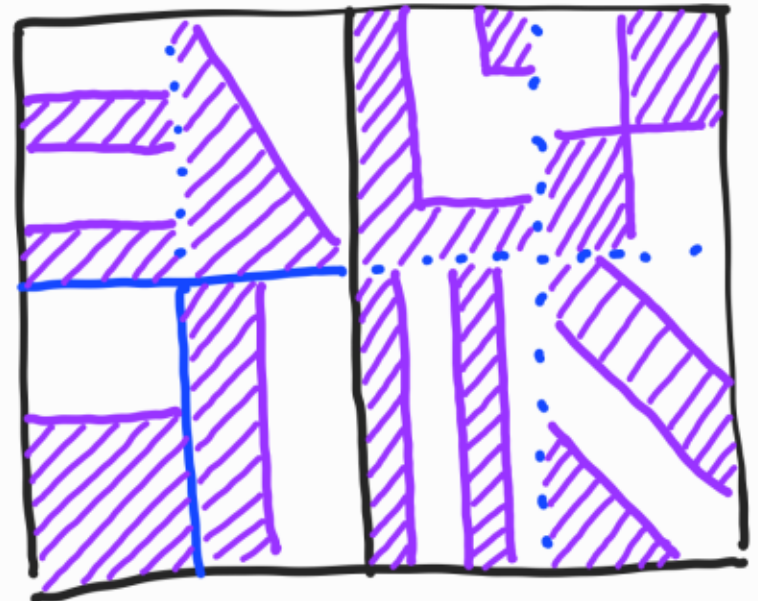
cosets of $H_i = \{H_i + u, u \in U_i\}$, $U_i \sim \mathbb{F}_p^{D_i}$

Pick codim-1 subspace

in each $H_i + u$: $r_{u,i}^T x = 0$

Lemma

There are $8d$ vectors in $\mathbb{F}_2^d$
s.t. any $\frac{3}{4}$ -proportion spans $\mathbb{F}_2^d$.



A Lower Bound for the Strong Arithmetic Regularity Lemma

$$\mathbb{F}_2^n = H_0 \supseteq H_1 \supseteq \dots \supseteq H_s$$

$\text{codim } H_i = D_i, 2^{D_i} = 8d_{i+1}$ cosets

Pick U_i s.t. $H_i \oplus U_i = \mathbb{F}_p^n$

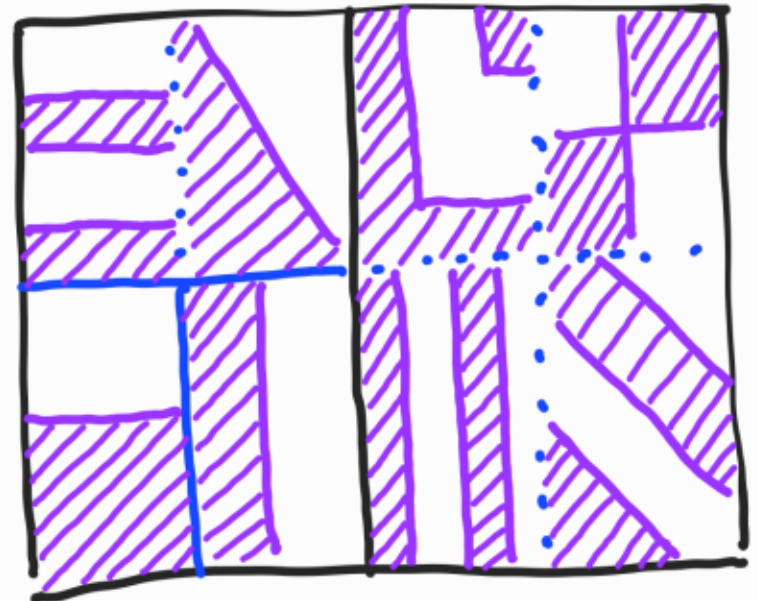
cosets of $H_i = \{H_i + u, u \in U_i\}, U_i \sim \mathbb{F}_p^{D_i}$

Pick codim-1 subspace
in each $H_i + u$: $r_{u,i}^T x = 0$

$$A_i = \text{[shaded rectangle]}$$

Lemma

There are $8d$ vectors in $\mathbb{F}_2^d$
s.t. any $\frac{3}{4}$ -proportion spans $\mathbb{F}_2^d$.



A Lower Bound for the Strong Arithmetic Regularity Lemma

$$\mathbb{F}_2^n = H_0 \supseteq H_1 \supseteq \dots \supseteq H_s$$

$\text{codim } H_i = D_i, 2^{D_i} = 8d_{i+1}$ cosets

Pick U_i s.t. $H_i \oplus U_i = \mathbb{F}_p^n$

cosets of $H_i = \{H_i + u, u \in U_i\}, U_i \sim \mathbb{F}_p^{D_i}$

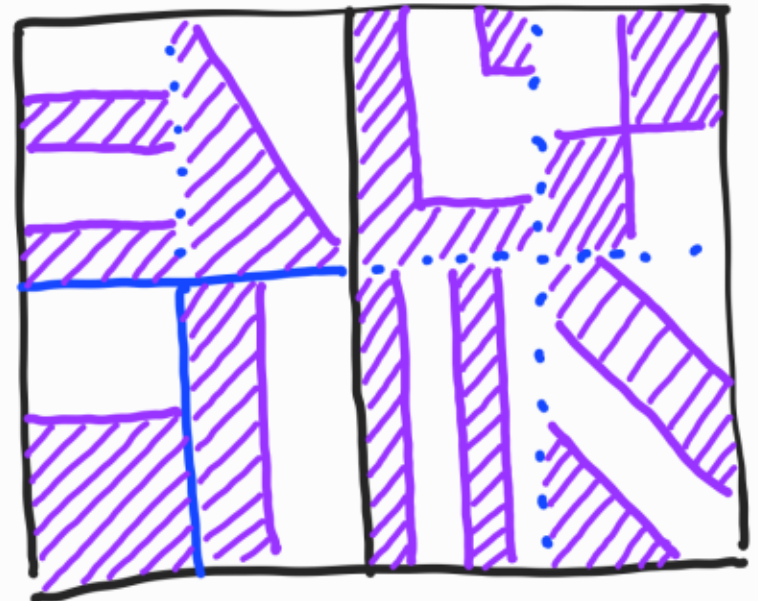
Pick codim-1 subspace
in each $H_i + u$: $r_{u,i}^T x = 0$

$$A_i = \text{[shaded rectangle]}$$

$$\{r_{u,i} \in H_i : u \in U_i\}$$

Lemma

There are $8d$ vectors in $\mathbb{F}_2^d$
s.t. any $\frac{3}{4}$ -proportion spans $\mathbb{F}_2^d$.



A Lower Bound for the Strong Arithmetic Regularity Lemma

$$\mathbb{F}_2^n = H_0 \supseteq H_1 \supseteq \dots \supseteq H_s$$

$\text{codim } H_i = D_i, 2^{D_i} = 8d_{i+1}$ cosets

Pick U_i s.t. $H_i \oplus U_i = \mathbb{F}_p^n$

cosets of $H_i = \{H_i + u, u \in U_i\}, U_i \sim \mathbb{F}_p^{D_i}$

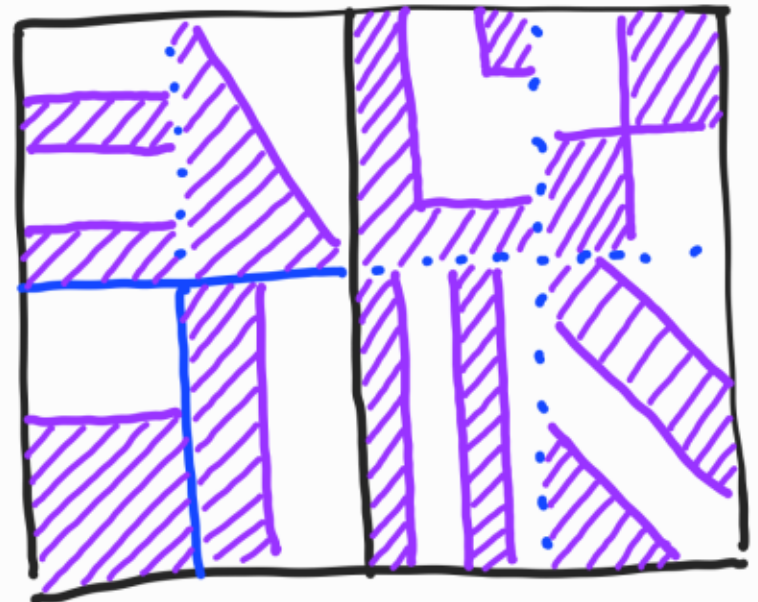
Pick codim-1 subspace
in each $H_i + u$: $r_{u,i}^T x = 0$

$$A_i = \text{[shaded rectangle]}$$

$\underbrace{\{r_{u,i} \in H_i : u \in U_i\}}_{8d_{i+1} \text{ vectors}}$

Lemma

There are $8d$ vectors in $\mathbb{F}_2^d$
s.t. any $\frac{3}{4}$ -proportion spans $\mathbb{F}_2^d$.



A Lower Bound for the Strong Arithmetic Regularity Lemma

$$\mathbb{F}_2^n = H_0 \supseteq H_1 \supseteq \dots \supseteq H_s$$

$\text{codim } H_i = D_i, 2^{D_i} = 8d_{i+1}$ cosets

Pick U_i s.t. $H_i \oplus U_i = \mathbb{F}_p^n$

cosets of $H_i = \{H_i + u, u \in U_i\}, U_i \sim \mathbb{F}_p^{D_i}$

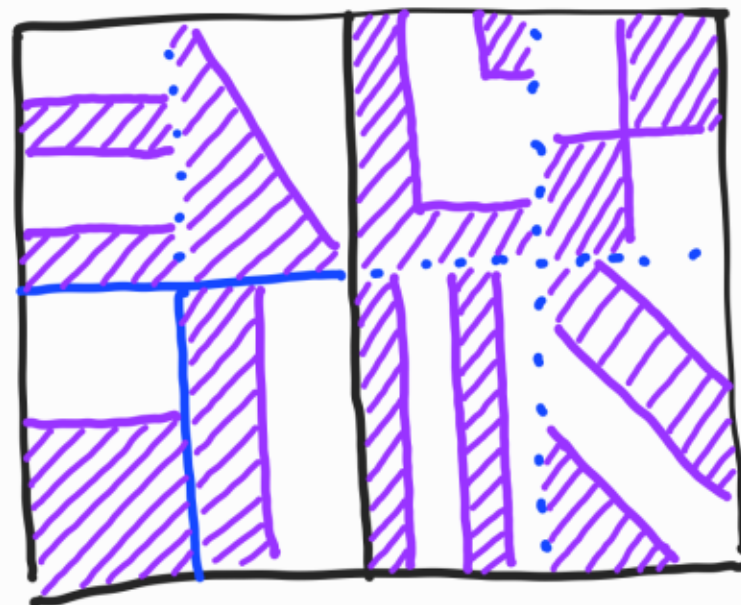
Pick codim-1 subspace
in each $H_i + u$: $r_{u,i}^T x = 0$

$$A_i = \text{[shaded rectangle]}$$

$\underbrace{\{r_{u,i} \in H_i : u \in U_i\}}_{8d_{i+1} \text{ vectors}} - \text{Lemma applied to } H_i / H_{i+1}$

Lemma

There are $8d$ vectors in $\mathbb{F}_2^d$
s.t. any $\frac{3}{4}$ -proportion spans $\mathbb{F}_2^d$.

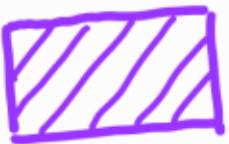


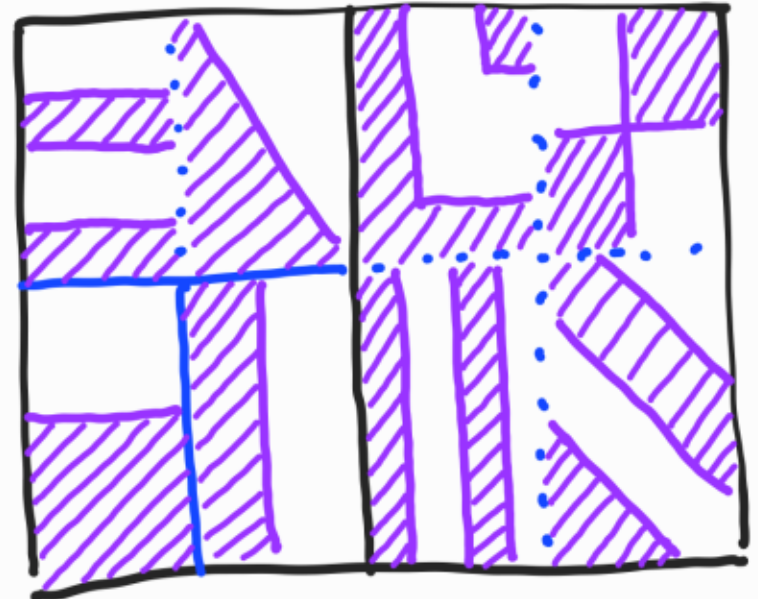
A Lower Bound for the Strong Arithmetic Regularity Lemma

$\mathbb{F}_2^n = H_0 \supseteq H_1 \supseteq \dots \supseteq H_s$
codim $H_i = D_i$, $2^{D_i} = 8d_{i+1}$ cosets

Lemma

There are $8d$ vectors in $\mathbb{F}_2^d$
s.t. any $\frac{3}{4}$ -proportion spans $\mathbb{F}_2^d$.

$A_i =$ 



A Lower Bound for the Strong Arithmetic Regularity Lemma

$$\mathbb{F}_2^n = H_0 \supseteq H_1 \supseteq \dots \supseteq H_s$$

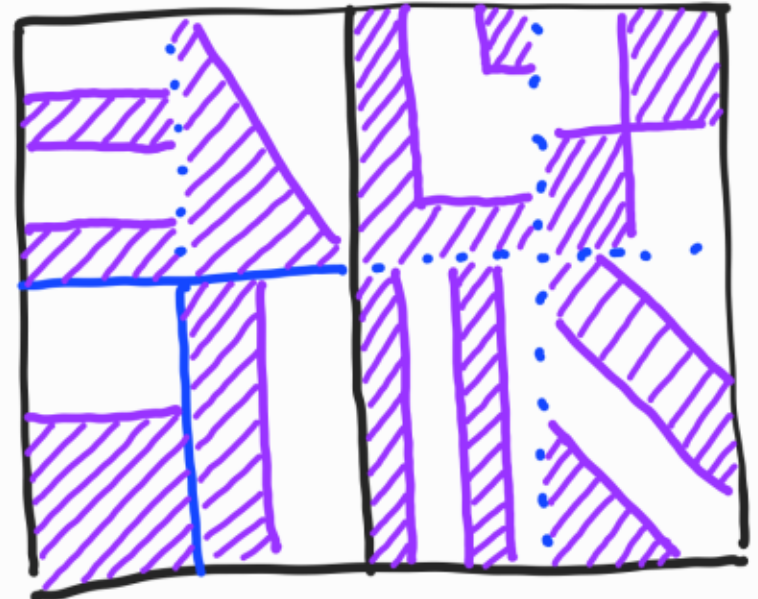
codim $H_i = D_i$, $2^{D_i} = 8d_{i+1}$ cosets

$$f(x) = \sum_{i=1}^s w_i \mathbb{1}_{A_i}(x), \quad \sum_{i=1}^s w_i \leq 1$$

Lemma

There are $8d$ vectors in $\mathbb{F}_2^d$ s.t. any $\frac{3}{4}$ -proportion spans $\mathbb{F}_2^d$.

$$A_i = \text{[shaded rectangle]}$$



A Lower Bound for the Strong Arithmetic Regularity Lemma

$$\mathbb{F}_2^n = H_0 \supseteq H_1 \supseteq \dots \supseteq H_s$$

codim $H_i = D_i$, $2^{D_i} = 8d_{i+1}$ cosets

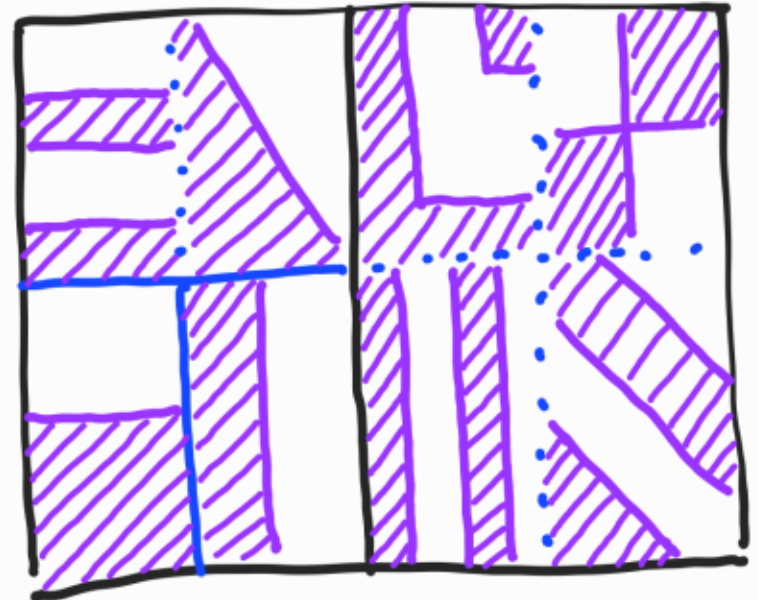
$$f(x) = \sum_{i=1}^s w_i \mathbb{1}_{A_i}(x), \quad \sum_{i=1}^s w_i \leq 1$$

$$A_i = \text{[shaded rectangle]}$$

Suppose ARL applied to f gives W .

Lemma

There are $8d$ vectors in $\mathbb{F}_2^d$ s.t. any $\frac{3}{4}$ -proportion spans $\mathbb{F}_2^d$.



A Lower Bound for the Strong Arithmetic Regularity Lemma

$$\mathbb{F}_2^n = H_0 \supseteq H_1 \supseteq \dots \supseteq H_s$$

$$\text{codim } H_i = D_i, 2^{D_i} = 8d_{i+1} \text{ cosets}$$

$$f(x) = \sum_{i=1}^s w_i \mathbb{1}_{A_i}(x), \quad \sum_{i=1}^s w_i \leq 1$$

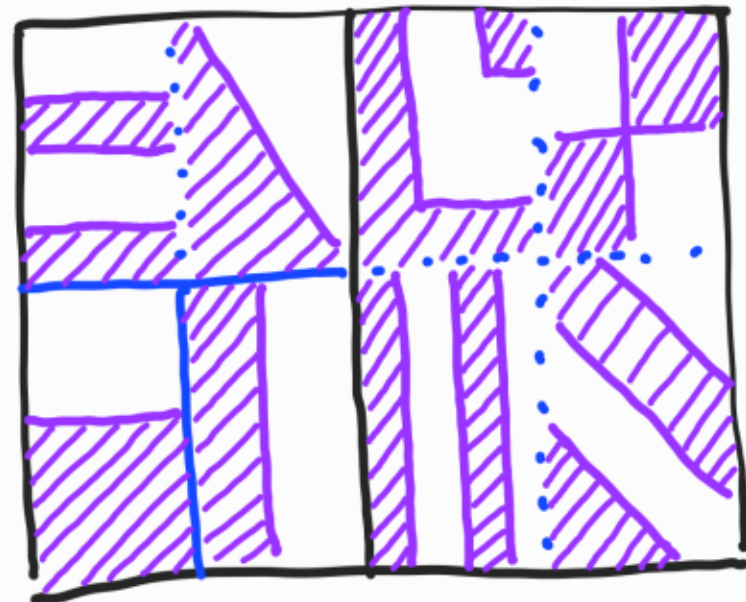
$$A_i = \text{[shaded rectangle]}$$

Suppose ARL applied to f gives W .

Goal: If $W \leq H_i$, then $W \leq H_{i+1}$

Lemma

There are $8d$ vectors in $\mathbb{F}_2^d$ s.t. any $\frac{3}{4}$ -proportion spans $\mathbb{F}_2^d$.



A Lower Bound for the Strong Arithmetic Regularity Lemma

$$\mathbb{F}_2^n = H_0 \supseteq H_1 \supseteq \dots \supseteq H_s$$

$\text{codim } H_i = D_i, 2^{D_i} = 8d_{i+1} \text{ cosets}$

$$f(x) = \sum_{i=1}^s w_i \mathbb{1}_{A_i}(x), \quad \sum_{i=1}^s w_i \leq 1$$

$$A_i = \text{[shaded rectangle]}$$

Suppose ARL applied to f gives W .

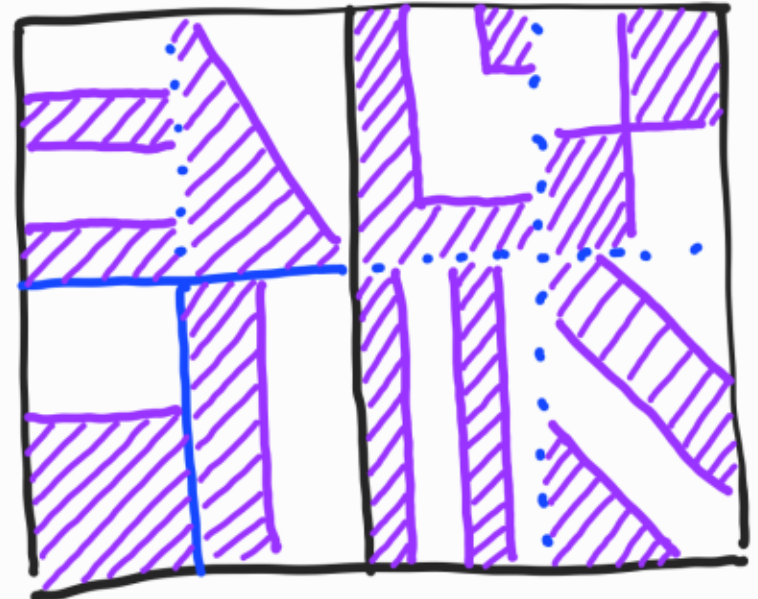
Goal: If $W \leq H_i$, then $W \leq H_{i+1}$

$$\Rightarrow W \leq H_s$$

$$\text{codim} \geq \text{twr}(s-1)$$

Lemma

There are $8d$ vectors in $\mathbb{F}_2^d$ s.t. any $\frac{3}{4}$ -proportion spans $\mathbb{F}_2^d$.



A Lower Bound for the Strong Arithmetic Regularity Lemma

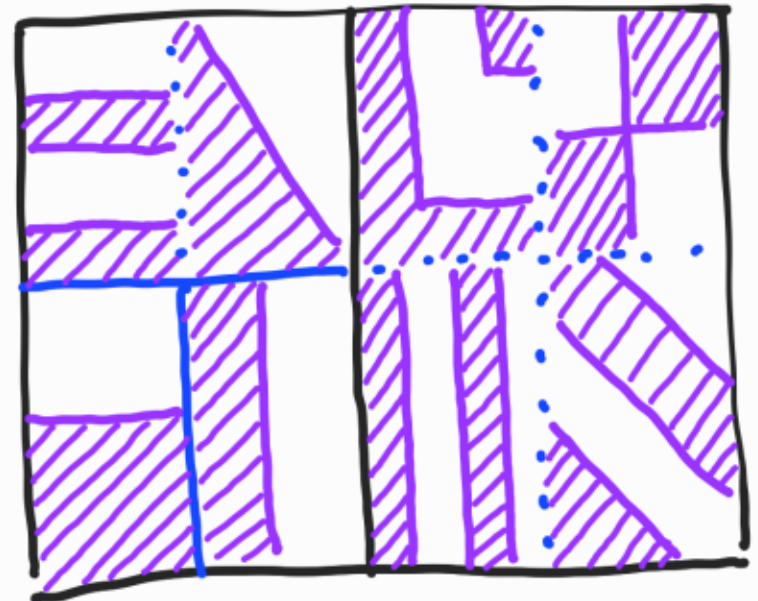
Suppose ARL applied to f gives W .

Goal: If $W \leq H_i$, then $W \leq H_{i+1}$

Lemma

There are $8d$ vectors in $\mathbb{F}_2^d$ s.t. any $\frac{3}{4}$ -proportion spans $\mathbb{F}_2^d$.

$\underbrace{\{r_{u,i} \in H_i : u \in U_i\}}_{8d_{i+1} \text{ vectors}}$ spans H_i/H_{i+1}



A Lower Bound for the Strong Arithmetic Regularity Lemma

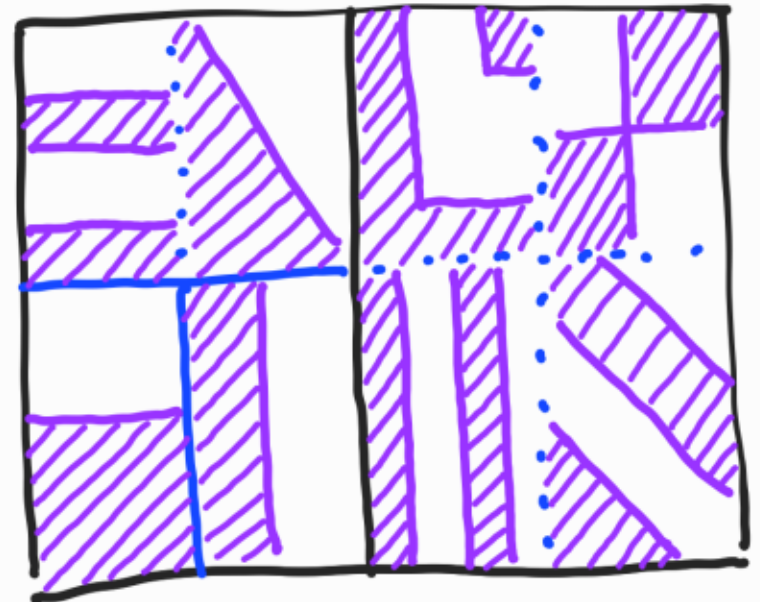
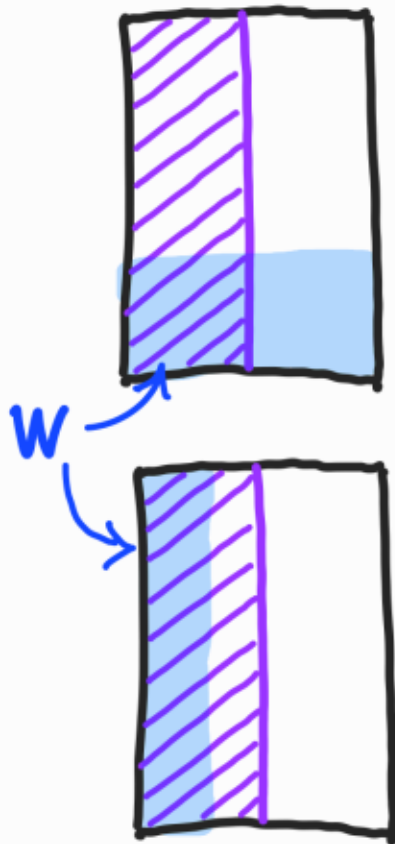
Suppose ARL applied to f gives W .

Goal: If $W \leq H_i$, then $W \leq H_{i+1}$

Lemma

There are $8d$ vectors in $\mathbb{F}_2^d$ s.t. any $\frac{3}{4}$ -proportion spans $\mathbb{F}_2^d$.

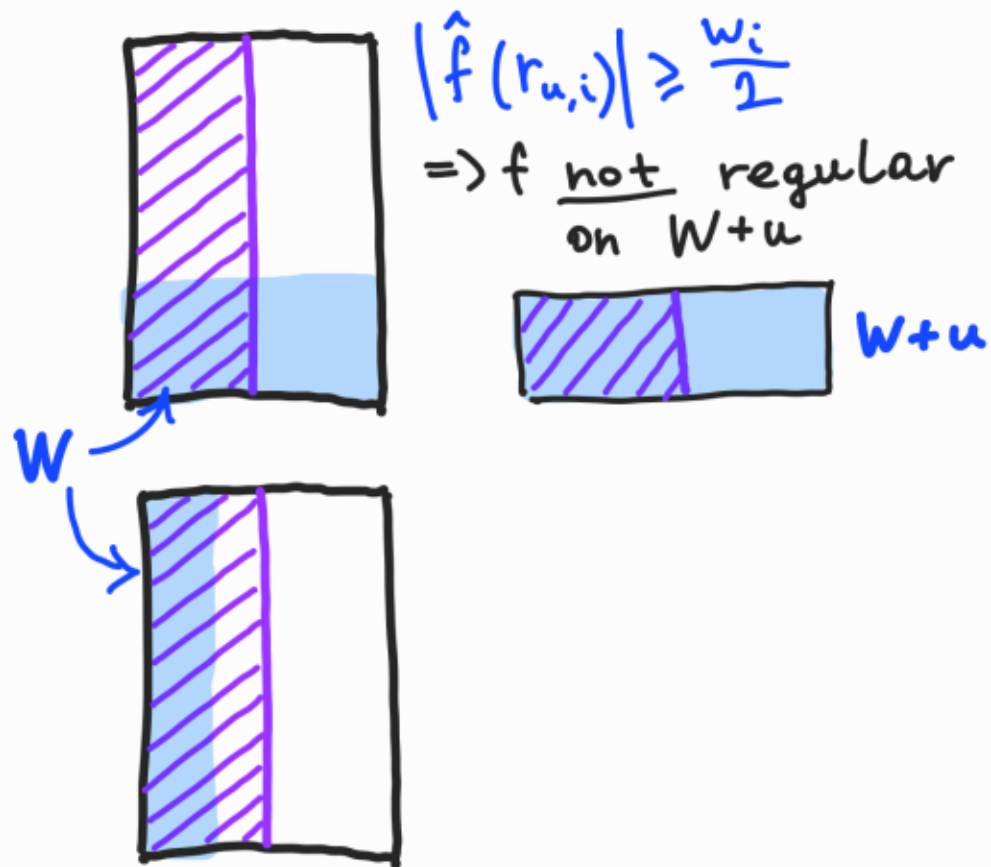
$\underbrace{\{r_{u,i} \in H_i : u \in U_i\}}_{8d_{i+1} \text{ vectors}} \text{ spans } H_i/H_{i+1}$



A Lower Bound for the Strong Arithmetic Regularity Lemma

Suppose ARL applied to f gives W .

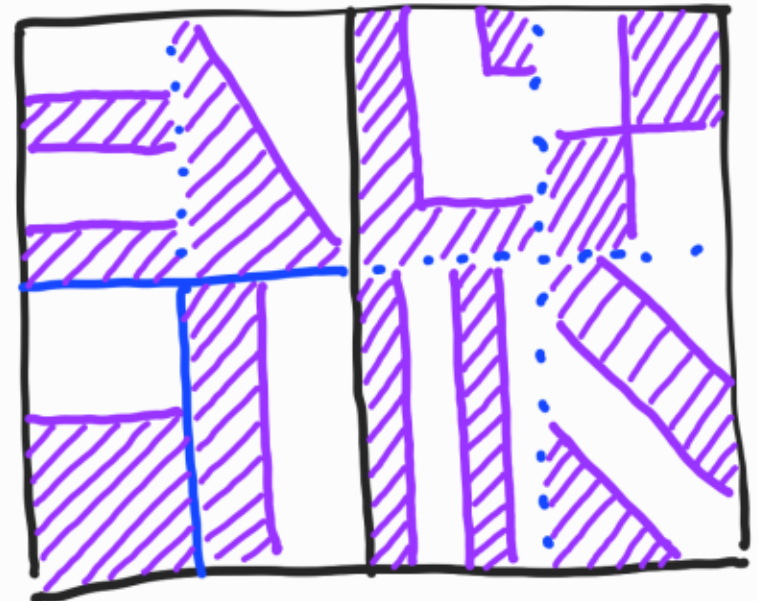
Goal: If $W \leq H_i$, then $W \leq H_{i+1}$



Lemma

There are $8d$ vectors in $\mathbb{F}_2^d$ s.t. any $\frac{3}{4}$ -proportion spans $\mathbb{F}_2^d$.

$\{r_{u,i} \in H_i : u \in U_i\}$ spans H_i/H_{i+1}
 $8d_{i+1}$ vectors



A Lower Bound for the Strong Arithmetic Regularity Lemma

Suppose ARL applied to f gives W .

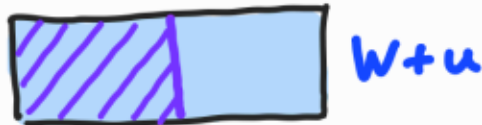
Goal: If $W \leq H_i$, then $W \leq H_{i+1}$

Lemma

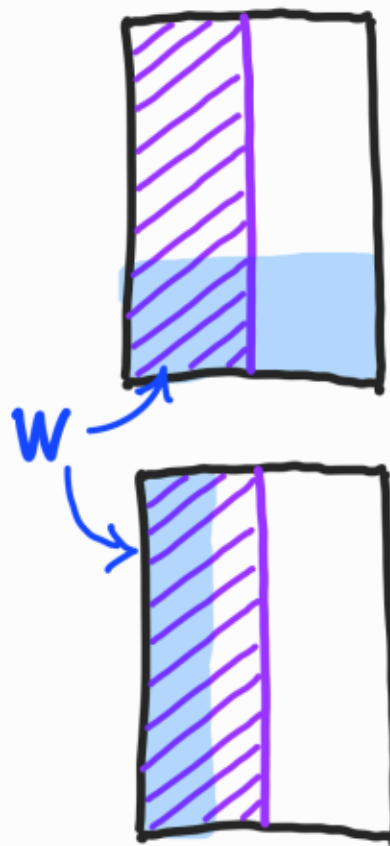
There are $8d$ vectors in $\mathbb{F}_2^d$ s.t. any $\frac{3}{4}$ -proportion spans $\mathbb{F}_2^d$.

$$|\hat{f}(r_{u,i})| \geq \frac{w_i}{2}$$

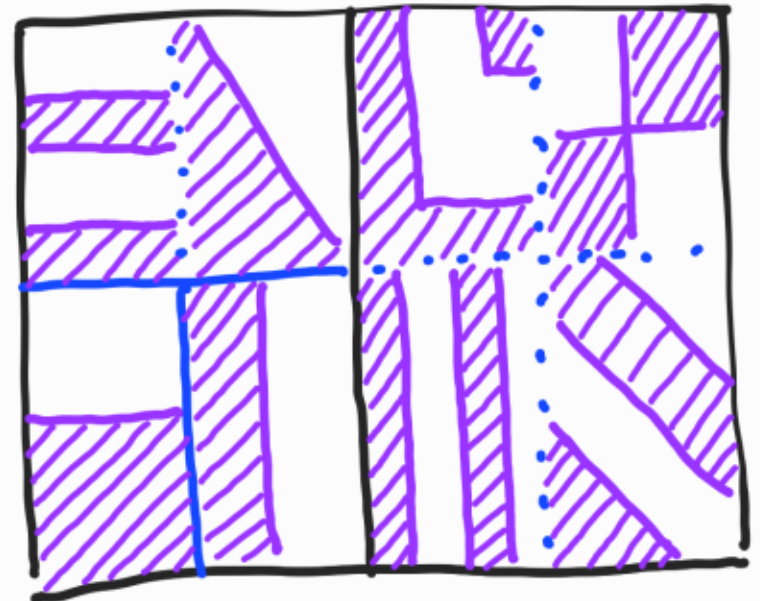
$$\Rightarrow f \text{ not regular on } W+u$$



$\{r_{u,i} \in H_i : u \in U_i\}$ spans H_i/H_{i+1}
 $8d_{i+1}$ vectors



$W \subseteq r_{u,i}^\perp$ for $(1-\varepsilon)$ -proportion of $\{r_{u,i}\}$

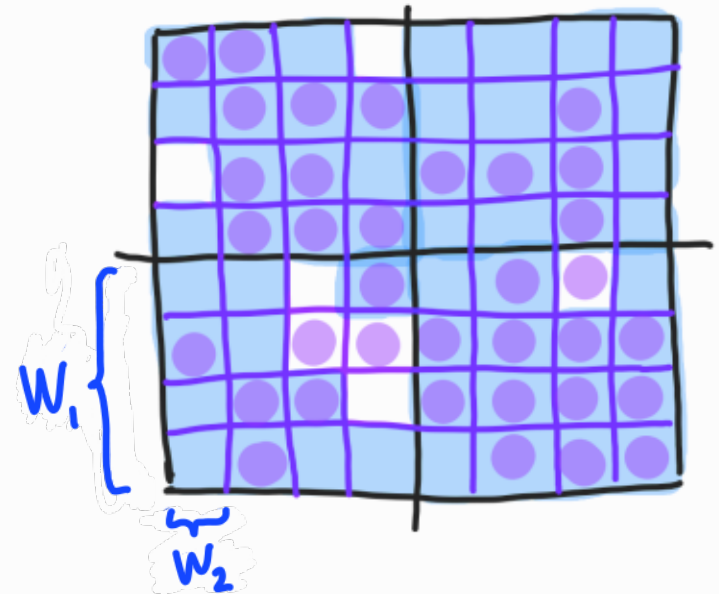


A Lower Bound for the Strong Arithmetic Regularity Lemma

Strong ARL

For every $\varepsilon: \mathbb{N} \rightarrow (0, 1)$ and $f: \mathbb{F}_2^n \rightarrow [0, 1]$, there are subspaces $W_2 \leq W_1 \leq \mathbb{F}_2^n$ of bounded codimensions $D_2 \geq D_1$ such that:

- i) f is $\varepsilon(D_1)$ -regular on all but an $\varepsilon(D_1)$ -proportion of cosets of W_2 ;
- ii) $\mathbb{E}(W_2) - \mathbb{E}(W_1) \leq \varepsilon(0)$.

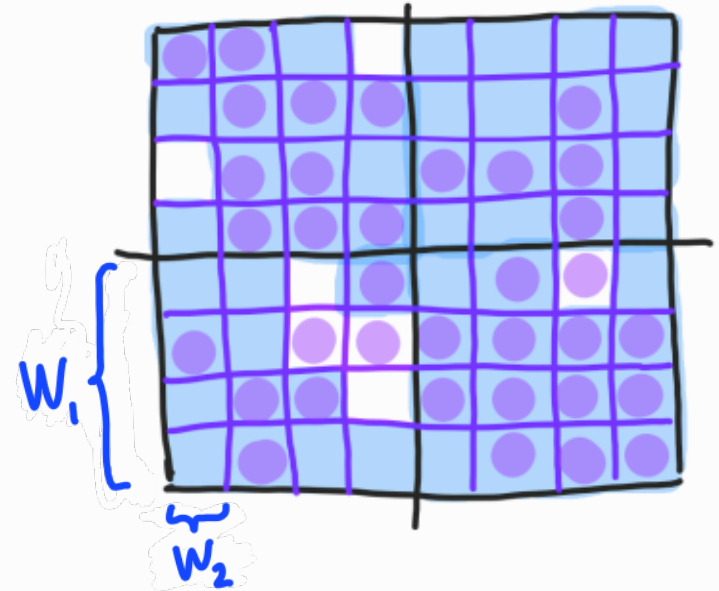


A Lower Bound for the Strong Arithmetic Regularity Lemma

Strong ARL

For every $\varepsilon: \mathbb{N} \rightarrow (0, 1)$ and $f: \mathbb{F}_2^n \rightarrow [0, 1]$, there are subspaces $W_2 \leq W_1 \leq \mathbb{F}_2^n$ of bounded codimensions $D_2 \geq D_1$ such that:

- i) f is $\varepsilon(D_1)$ -regular on all but an $\varepsilon(D_1)$ -proportion of cosets of W_2 ;
- ii) $\mathcal{E}(W_2) - \mathcal{E}(W_1) \leq \varepsilon(0)$.



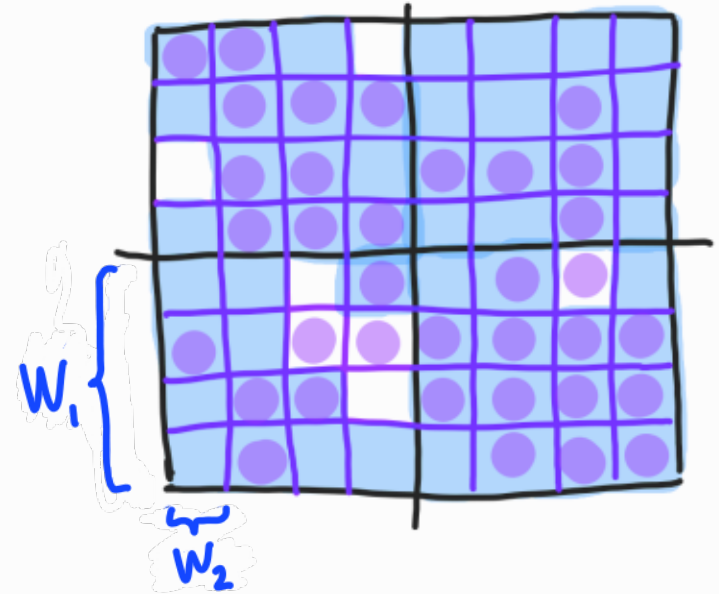
$$\mathbb{F}_2^n = \underbrace{H_0^{(1)} \geq H_1^{(1)} \geq \dots \geq H_{s_1}^{(1)}}_{W_1} \geq \underbrace{H_1^{(2)} \geq \dots \geq H_{s_2}^{(2)}}_{W_2} \geq \underbrace{H_1^{(3)} \geq \dots \geq H_{s_3}^{(3)}}_{W_3} \geq \dots \geq \underbrace{H_1^{(t)} \geq \dots \geq H_{s_t}^{(t)}}_{W_t}$$

A Lower Bound for the Strong Arithmetic Regularity Lemma

Strong ARL

For every $\varepsilon: \mathbb{N} \rightarrow (0, 1)$ and $f: \mathbb{F}_2^n \rightarrow [0, 1]$, there are subspaces $W_2 \leq W_1 \leq \mathbb{F}_2^n$ of bounded codimensions $D_2 \geq D_1$ such that:

- i) f is $\varepsilon(D_1)$ -regular on all but an $\varepsilon(D_1)$ -proportion of cosets of W_2 ;
- ii) $\mathcal{E}(W_2) - \mathcal{E}(W_1) \leq \varepsilon(0)$.



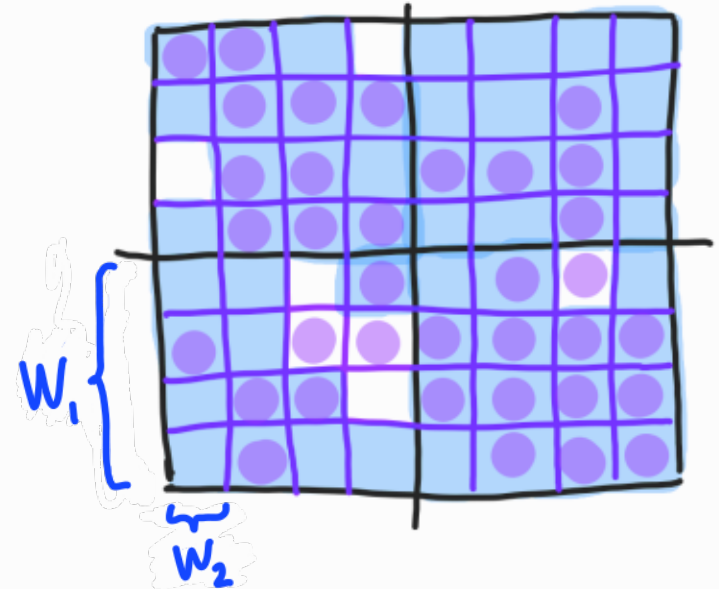
$$\mathbb{F}_2^n = \underbrace{H_0^{(1)} \geq H_1^{(1)} \geq \dots \geq H_{s_1}^{(1)}}_{\varepsilon(0)\text{-regular}} \geq \underbrace{H_1^{(2)} \geq \dots \geq H_{s_2}^{(2)}} \geq \underbrace{H_1^{(3)} \geq \dots \geq H_{s_3}^{(3)}} \geq \dots \geq \underbrace{H_1^{(t)} \geq \dots \geq H_{s_t}^{(t)}}$$

A Lower Bound for the Strong Arithmetic Regularity Lemma

Strong ARL

For every $\varepsilon: \mathbb{N} \rightarrow (0, 1)$ and $f: \mathbb{F}_2^n \rightarrow [0, 1]$, there are subspaces $W_2 \leq W_1 \leq \mathbb{F}_2^n$ of bounded codimensions $D_2 \geq D_1$ such that:

- i) f is $\varepsilon(D_1)$ -regular on all but an $\varepsilon(D_1)$ -proportion of cosets of W_2 ;
- ii) $\mathcal{E}(W_2) - \mathcal{E}(W_1) \leq \varepsilon(0)$.



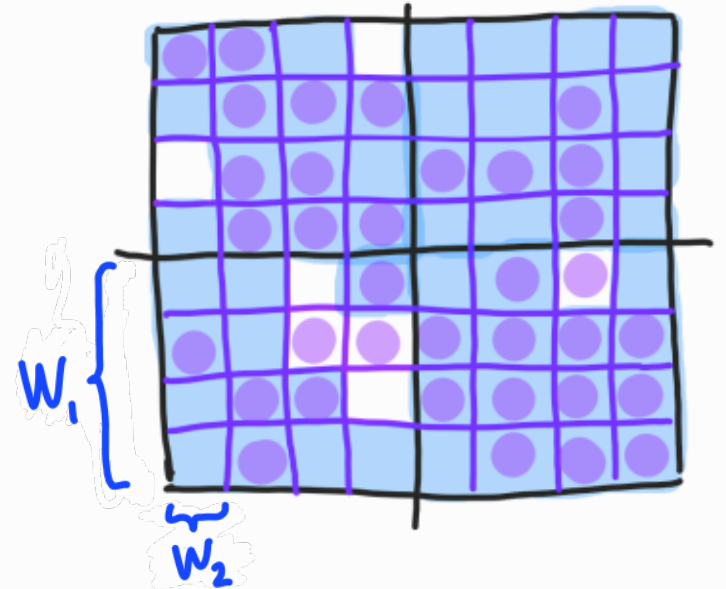
$$\mathbb{F}_2^n = \underbrace{H_0^{(1)} \geq H_1^{(1)} \geq \dots \geq H_{s_1}^{(1)}}_{\varepsilon(0)\text{-regular}} \geq \underbrace{H_1^{(2)} \geq \dots \geq H_{s_2}^{(2)}}_{\varepsilon(D_1^{(1)})\text{-regular}} \geq \underbrace{H_1^{(3)} \geq \dots \geq H_{s_3}^{(3)}}_{\dots} \geq \dots \geq \underbrace{H_1^{(t)} \geq \dots \geq H_{s_t}^{(t)}}_{\dots}$$

A Lower Bound for the Strong Arithmetic Regularity Lemma

Strong ARL

For every $\varepsilon: \mathbb{N} \rightarrow (0, 1)$ and $f: \mathbb{F}_2^n \rightarrow [0, 1]$, there are subspaces $W_2 \leq W_1 \leq \mathbb{F}_2^n$ of bounded codimensions $D_2 \geq D_1$ such that:

- i) f is $\varepsilon(D_1)$ -regular on all but an $\varepsilon(D_1)$ -proportion of cosets of W_2 ;
- ii) $\mathcal{E}(W_2) - \mathcal{E}(W_1) \leq \varepsilon(0)$.



$$\mathbb{F}_2^n = \underbrace{H_0^{(1)} \geq H_1^{(1)} \geq \dots \geq H_{s_1}^{(1)}}_{\varepsilon(0)\text{-regular}} \geq \underbrace{H_1^{(2)} \geq \dots \geq H_{s_2}^{(2)}}_{\varepsilon(D_1^{(1)})\text{-regular}} \geq \underbrace{H_1^{(3)} \geq \dots \geq H_{s_3}^{(3)}}_{\varepsilon(D_1^{(2)})\text{-regular}} \geq \dots \geq \underbrace{H_1^{(t)} \geq \dots \geq H_{s_t}^{(t)}}_{\varepsilon(D_1^{(t-1)})\text{-regular}}$$

A Lower Bound for the Strong Arithmetic Regularity Lemma

$$\left[\begin{array}{l|l} W_2 \leq W_1 & f \text{ is } \varepsilon(D_1)\text{-regular on (almost all) cosets of } W_2; \\ \hline \underbrace{D_2 \geq D_1}_{\text{codimensions}} & \varepsilon(W_2) - \varepsilon(W_1) \leq \varepsilon(0) \end{array} \right] \quad \text{Strong ARL}$$

$$\mathbb{A}_2^n = \underbrace{H_0^{(1)} \geq H_1^{(1)} \geq \dots \geq H_{s_1}^{(1)}}_{\varepsilon(0)\text{-regular}} \underbrace{\geq H_1^{(2)} \geq \dots \geq H_{s_2}^{(2)}}_{\varepsilon(D^{(1)})\text{-regular}} \underbrace{\geq H_1^{(3)} \geq \dots \geq H_{s_3}^{(3)}}_{\varepsilon(D^{(2)})\text{-regular}} \underbrace{\geq H_1^{(t)} \geq \dots \geq H_{s_t}^{(t)}}_{\varepsilon(D^{(t-1)})\text{-regular}}$$

A Lower Bound for the Strong Arithmetic Regularity Lemma

$W_2 \leq W_1$ $\underbrace{D_2 \geq D_1}_{\text{codimensions}}$	f is $\varepsilon(D_1)$ -regular on (almost all) cosets of W_2 ; $\varepsilon(W_2) - \varepsilon(W_1) \leq \varepsilon(0)$	Strong ARL
---	---	-------------------

$$\mathbb{F}_2^n = \underbrace{H_0^{(1)} \geq H_1^{(1)} \geq \dots \geq H_{s_1}^{(1)}}_{\varepsilon(0)\text{-regular}} \underbrace{\geq H_1^{(2)} \geq \dots \geq H_{s_2}^{(2)}}_{\varepsilon(D_1^{(1)})\text{-regular}} \underbrace{\geq H_1^{(3)} \geq \dots \geq H_{s_3}^{(3)}}_{\varepsilon(D_2^{(1)})\text{-regular}} \underbrace{\geq H_1^{(t)} \geq \dots \geq H_{s_t}^{(t)}}_{\varepsilon(D_{t-1}^{(1)})\text{-regular}}$$

$$D_1 \geq 0 \Rightarrow W_2 \leq H_{s_1}^{(1)}$$

Since f is $\varepsilon(0)$ -regular on W_2 -cosets

A Lower Bound for the Strong Arithmetic Regularity Lemma

$W_2 \leq W_1$ $\underbrace{D_2 \geq D_1}_{\text{codimensions}}$	f is $\varepsilon(D_1)$ -regular on (almost all) cosets of W_2 ; $\varepsilon(W_2) - \varepsilon(W_1) \leq \varepsilon(0)$	Strong ARL
---	---	-------------------

$$\mathbb{A}_2^n = \underbrace{H_0^{(1)} \geq H_1^{(1)} \geq \dots \geq H_{s_1}^{(1)}}_{\varepsilon(0)\text{-regular}} \underbrace{\geq H_1^{(2)} \geq \dots \geq H_{s_2}^{(2)}}_{\varepsilon(D^{(1)})\text{-regular}} \underbrace{\geq H_1^{(3)} \geq \dots \geq H_{s_3}^{(3)}}_{\varepsilon(D^{(2)})\text{-regular}} \dots \underbrace{\geq H_1^{(t)} \geq \dots \geq H_{s_t}^{(t)}}_{\varepsilon(D^{(t-1)})\text{-regular}}$$

$D_1 \geq 0 \Rightarrow W_2 \leq H_{s_1}^{(1)}$

$D_1 \geq D^{(1)} \Rightarrow W_2 \leq H_{s_2}^{(2)}$

Since f is $\varepsilon(D^{(1)})$ -regular on W_2 -cosets

A Lower Bound for the Strong Arithmetic Regularity Lemma

$W_2 \leq W_1$ $\underbrace{D_2 \geq D_1}_{\text{codimensions}}$	f is $\varepsilon(D_1)$ -regular on (almost all) cosets of W_2 ; $\varepsilon(W_2) - \varepsilon(W_1) \leq \varepsilon(0)$	Strong ARL
---	---	-------------------

$$\begin{array}{ccccccc}
 \mathbb{A}_2^n = & \underbrace{H_0^{(1)} \geq H_1^{(1)} \geq \dots \geq H_{s_1}^{(1)}}_{\substack{\varepsilon(0)\text{-regular} \\ D_1 \geq 0 \Rightarrow W_2 \leq H_{s_1}^{(1)}}} & \underbrace{H_1^{(2)} \geq \dots \geq H_{s_2}^{(2)}}_{\substack{\varepsilon(D_1^{(1)})\text{-regular} \\ D_1 \geq D_1^{(1)} \Rightarrow W_2 \leq H_{s_2}^{(2)}}} & \underbrace{H_1^{(3)} \geq \dots \geq H_{s_3}^{(3)}}_{\substack{\varepsilon(D_1^{(2)})\text{-regular} \\ D_1 \geq D_1^{(2)} \Rightarrow W_2 \leq H_{s_3}^{(3)}}} & \dots & \underbrace{H_1^{(t)} \geq \dots \geq H_{s_t}^{(t)}}_{\substack{\varepsilon(D_1^{(t-1)})\text{-regular} \\ D_1 \geq D_1^{(t-1)} \Rightarrow W_2 \leq H_{s_t}^{(t)}}}
 \end{array}$$

A Lower Bound for the Strong Arithmetic Regularity Lemma

$W_2 \leq W_1$ $\underbrace{D_2 \geq D_1}_{\text{codimensions}}$	f is $\varepsilon(D_1)$ -regular on (almost all) cosets of W_2 ; $\varepsilon(W_2) - \varepsilon(W_1) \leq \varepsilon(0)$	Strong ARL
---	---	------------

$$\begin{aligned}
 \mathbb{F}_2^n = & \underbrace{H_0^{(1)} \geq H_1^{(1)} \geq \dots \geq H_{s_1}^{(1)}}_{\substack{\varepsilon(0)\text{-regular} \\ D_1 \geq 0 \Rightarrow W_2 \leq H_{s_1}^{(1)}}} \geq \underbrace{H_1^{(2)} \geq \dots \geq H_{s_2}^{(2)}}_{\substack{\varepsilon(D_1^{(1)})\text{-regular} \\ D_1 \geq D_1^{(1)} \Rightarrow W_2 \leq H_{s_2}^{(2)}}} \geq \underbrace{H_1^{(3)} \geq \dots \geq H_{s_3}^{(3)}}_{\substack{\varepsilon(D_1^{(2)})\text{-regular} \\ D_1 \geq D_1^{(2)} \Rightarrow W_2 \leq H_{s_3}^{(3)}}} \geq \dots \geq \underbrace{H_1^{(t)} \geq \dots \geq H_{s_t}^{(t)}}_{\substack{\varepsilon(D_1^{(t-1)})\text{-regular} \\ D_1 \geq D_1^{(t-1)} \Rightarrow W_2 \leq H_{s_t}^{(t)}}} \\
 & \hspace{15em} \text{Done } \checkmark
 \end{aligned}$$

A Lower Bound for the Strong Arithmetic Regularity Lemma

$W_2 \leq W_1$ $\underbrace{D_2 \geq D_1}_{\text{codimensions}}$	f is $\varepsilon(D_1)$ -regular on (almost all) cosets of W_2 ; $\varepsilon(W_2) - \varepsilon(W_1) \leq \varepsilon(0)$	Strong ARL
---	---	-------------------

$$\begin{array}{ccccccc}
 \mathbb{F}_2^n = & \underbrace{H_0^{(1)} \geq H_1^{(1)} \geq \dots \geq H_{s_1}^{(1)}}_{\substack{\varepsilon(0)\text{-regular} \\ D_1 \geq 0 \Rightarrow W_2 \leq H_{s_1}^{(1)}}} & \underbrace{H_1^{(2)} \geq \dots \geq H_{s_2}^{(2)}}_{\substack{\varepsilon(D_1^{(1)})\text{-regular} \\ D_1 \geq D^{(1)} \Rightarrow W_2 \leq H_{s_2}^{(2)}}} & \underbrace{H_1^{(3)} \geq \dots \geq H_{s_3}^{(3)}}_{\substack{\varepsilon(D_1^{(2)})\text{-regular} \\ D_1 \geq D^{(2)} \Rightarrow W_2 \leq H_{s_3}^{(3)}}} & \dots & \underbrace{H_1^{(t)} \geq \dots \geq H_{s_t}^{(t)}}_{\substack{\varepsilon(D_1^{(t-1)})\text{-regular} \\ D_1 \geq D^{(t-1)} \Rightarrow W_2 \leq H_{s_t}^{(t)}}}
 \end{array}$$

Assume $D_1 < D^{(i)}$
 for some $i < t-1$

A Lower Bound for the Strong Arithmetic Regularity Lemma

$W_2 \leq W_1$ $\underbrace{D_2 \geq D_1}_{\text{codimensions}}$	f is $\varepsilon(D_1)$ -regular on (almost all) cosets of W_2 ; $\varepsilon(W_2) - \varepsilon(W_1) \leq \varepsilon(0)$	Strong ARL
---	---	-------------------

$$\mathbb{A}_2^n = \underbrace{H_0^{(1)} \geq H_1^{(1)} \geq \dots \geq H_{s_1}^{(1)}}_{\substack{\varepsilon(0)\text{-regular} \\ D_1 \geq 0 \Rightarrow W_2 \leq H_{s_1}^{(1)}}} \underbrace{H_1^{(2)} \geq \dots \geq H_{s_2}^{(2)}}_{\substack{\varepsilon(D_1^{(1)})\text{-regular} \\ D_1 \geq D^{(1)} \Rightarrow W_2 \leq H_{s_2}^{(2)}}} \underbrace{H_1^{(3)} \geq \dots \geq H_{s_3}^{(3)}}_{\substack{\varepsilon(D_1^{(2)})\text{-regular} \\ D_1 \geq D^{(2)} \Rightarrow W_2 \leq H_{s_3}^{(3)}}} \dots \underbrace{H_1^{(t)} \geq \dots \geq H_{s_t}^{(t)}}_{\substack{\varepsilon(D_1^{(t-1)})\text{-regular} \\ D_1 \geq D^{(t-1)} \Rightarrow W_2 \leq H_{s_t}^{(t)}}}$$

Assume $D_1 < D^{(i)}$
 for some $i < t-1 \Rightarrow H_1^{(i)} \geq \dots \geq H_{s_{i-2}}^{(i)} \geq H_{s_{i-1}}^{(i)} \geq H_{s_i}^{(i)}$

A Lower Bound for the Strong Arithmetic Regularity Lemma

$W_2 \leq W_1$ $\underbrace{D_2 \geq D_1}_{\text{codimensions}}$	f is $\varepsilon(D_1)$ -regular on (almost all) cosets of W_2 ; $\varepsilon(W_2) - \varepsilon(W_1) \leq \varepsilon(0)$	Strong ARL
---	---	-------------------

$$\mathbb{A}_2^n = \underbrace{H_0^{(1)} \geq H_1^{(1)} \geq \dots \geq H_{s_1}^{(1)}}_{\substack{\varepsilon(0)\text{-regular} \\ D_1 \geq 0 \Rightarrow W_2 \leq H_{s_1}^{(1)}}} \underbrace{H_1^{(2)} \geq \dots \geq H_{s_2}^{(2)}}_{\substack{\varepsilon(D_1^{(1)})\text{-regular} \\ D_1 \geq D^{(1)} \Rightarrow W_2 \leq H_{s_2}^{(2)}}} \underbrace{H_1^{(3)} \geq \dots \geq H_{s_3}^{(3)}}_{\substack{\varepsilon(D_1^{(2)})\text{-regular} \\ D_1 \geq D^{(2)} \Rightarrow W_2 \leq H_{s_3}^{(3)}}} \dots \underbrace{H_1^{(t)} \geq \dots \geq H_{s_t}^{(t)}}_{\substack{\varepsilon(D_1^{(t-1)})\text{-regular} \\ D_1 \geq D^{(t-1)} \Rightarrow W_2 \leq H_{s_t}^{(t)}}}$$

Assume $D_1 < D^{(i)}$ for some $i < t-1 \Rightarrow H_1^{(i)} \geq \dots \geq H_{s_{i-2}}^{(i)} \geq H_{s_{i-1}}^{(i)} \geq H_{s_i}^{(i)} \geq W_2$

A Lower Bound for the Strong Arithmetic Regularity Lemma

$W_2 \leq W_1$ $\underbrace{D_2 \geq D_1}_{\text{codimensions}}$	f is $\varepsilon(D_1)$ -regular on (almost all) cosets of W_2 ; $\varepsilon(W_2) - \varepsilon(W_1) \leq \varepsilon(0)$	Strong ARL
---	---	-------------------

$$\mathbb{F}_2^n = \underbrace{H_0^{(1)} \geq H_1^{(1)} \geq \dots \geq H_{s_1}^{(1)}}_{\substack{\varepsilon(0)\text{-regular} \\ D_1 \geq 0 \Rightarrow W_2 \leq H_{s_1}^{(1)}}} \geq \underbrace{H_1^{(2)} \geq \dots \geq H_{s_2}^{(2)}}_{\substack{\varepsilon(D_1^{(1)})\text{-regular} \\ D_1 \geq D^{(1)} \Rightarrow W_2 \leq H_{s_2}^{(2)}}} \geq \underbrace{H_1^{(3)} \geq \dots \geq H_{s_3}^{(3)}}_{\substack{\varepsilon(D_1^{(2)})\text{-regular} \\ D_1 \geq D^{(2)} \Rightarrow W_2 \leq H_{s_3}^{(3)}}} \geq \dots \geq \underbrace{H_1^{(t)} \geq \dots \geq H_{s_t}^{(t)}}_{\substack{\varepsilon(D_1^{(t-1)})\text{-regular} \\ D_1 \geq D^{(t-1)} \Rightarrow W_2 \leq H_{s_t}^{(t)}}}$$

Assume $D_1 < D^{(i)}$ for some $i < t-1 \Rightarrow$

$$\underbrace{H_1^{(i)} \geq \dots \geq H_{s_{i-2}}^{(i)} \geq H_{s_{i-1}}^{(i)} \geq H_{s_i}^{(i)}}_{\text{ENERGY}} \geq W_2$$

A Lower Bound for the Strong Arithmetic Regularity Lemma

$W_2 \leq W_1$ $\underbrace{D_2 \geq D_1}_{\text{codimensions}}$	f is $\varepsilon(D_1)$ -regular on (almost all) cosets of W_2 ; $\mathcal{E}(W_2) - \mathcal{E}(W_1) \leq \varepsilon(0)$	Strong ARL
---	---	-------------------

$$\mathbb{F}_2^n = \underbrace{H_0^{(1)} \geq H_1^{(1)} \geq \dots \geq H_{s_1}^{(1)}}_{\substack{\varepsilon(0)\text{-regular} \\ D_1 \geq 0 \Rightarrow W_2 \leq H_{s_1}^{(1)}}} \underbrace{\geq H_1^{(2)} \geq \dots \geq H_{s_2}^{(2)}}_{\substack{\varepsilon(D_1^{(1)})\text{-regular} \\ D_1 \geq D^{(1)} \Rightarrow W_2 \leq H_{s_2}^{(2)}}} \underbrace{\geq H_1^{(3)} \geq \dots \geq H_{s_3}^{(3)}}_{\substack{\varepsilon(D_1^{(2)})\text{-regular} \\ D_1 \geq D^{(2)} \Rightarrow W_2 \leq H_{s_3}^{(3)}}} \dots \underbrace{\geq H_1^{(t)} \geq \dots \geq H_{s_t}^{(t)}}_{\substack{\varepsilon(D_1^{(t-1)})\text{-regular} \\ D_1 \geq D^{(t-1)} \Rightarrow W_2 \leq H_{s_t}^{(t)}}}$$

Assume $D_1 < D^{(i)}$ for some $i < t-1$

$\Rightarrow \mathcal{E}(W_1) \lesssim \mathcal{E}(H_{s_{i-2}}^{(i)})$

$$\underbrace{H_1^{(i)} \geq \dots \geq H_{s_{i-2}}^{(i)}}_{\text{ENERGY}} \geq H_{s_{i-1}}^{(i)} \geq H_{s_i}^{(i)} \geq W_2$$

A Lower Bound for the Strong Arithmetic Regularity Lemma

$W_2 \leq W_1$ $\underbrace{D_2 \geq D_1}_{\text{codimensions}}$	f is $\varepsilon(D_1)$ -regular on (almost all) cosets of W_2 ; $\mathcal{E}(W_2) - \mathcal{E}(W_1) \leq \varepsilon(0)$	Strong ARL
---	---	-------------------

$$\mathbb{F}_2^n = \underbrace{H_0^{(1)} \geq H_1^{(1)} \geq \dots \geq H_{s_1}^{(1)}}_{\substack{\varepsilon(0)\text{-regular} \\ D_1 \geq 0 \Rightarrow W_2 \leq H_{s_1}^{(1)}}} \underbrace{H_1^{(2)} \geq \dots \geq H_{s_2}^{(2)}}_{\substack{\varepsilon(D_1^{(1)})\text{-regular} \\ D_1 \geq D^{(1)} \Rightarrow W_2 \leq H_{s_2}^{(2)}}} \underbrace{H_1^{(3)} \geq \dots \geq H_{s_3}^{(3)}}_{\substack{\varepsilon(D_1^{(2)})\text{-regular} \\ D_1 \geq D^{(2)} \Rightarrow W_2 \leq H_{s_3}^{(3)}}} \dots \underbrace{H_1^{(t)} \geq \dots \geq H_{s_t}^{(t)}}_{\substack{\varepsilon(D_1^{(t-1)})\text{-regular} \\ D_1 \geq D^{(t-1)} \Rightarrow W_2 \leq H_{s_t}^{(t)}}}$$

Assume $D_1 < D^{(i)}$ for some $i < t-1$
 $\Rightarrow \mathcal{E}(W_1) \lesssim \mathcal{E}(H_{s_{i-2}}^{(i)})$

$$\Rightarrow \underbrace{H_1^{(i)} \geq \dots \geq H_{s_{i-2}}^{(i)}}_{\text{ENERGY}} \geq \underbrace{H_{s_{i-1}}^{(i)}}_{W_1} \geq H_{s_i}^{(i)} \geq W_2$$

A Lower Bound for the Strong Arithmetic Regularity Lemma

$$\left[\begin{array}{l} W_2 \leq W_1 \\ \underbrace{D_2 \geq D_1}_{\text{codimensions}} \end{array} \right] \left\{ \begin{array}{l} f \text{ is } \varepsilon(D_1)\text{-regular on (almost all) cosets of } W_2; \\ \mathcal{E}(W_2) - \mathcal{E}(W_1) \leq \varepsilon(0) \end{array} \right. \quad \text{Strong ARL}$$

$$\mathbb{F}_2^n = \underbrace{H_0^{(1)} \geq H_1^{(1)} \geq \dots \geq H_{s_1}^{(1)}}_{\substack{\varepsilon(0)\text{-regular} \\ D_1 \geq 0 \Rightarrow W_2 \leq H_{s_1}^{(1)}}} \underbrace{H_1^{(2)} \geq \dots \geq H_{s_2}^{(2)}}_{\substack{\varepsilon(D_1^{(1)})\text{-regular} \\ D_1 \geq D^{(1)} \Rightarrow W_2 \leq H_{s_2}^{(2)}}} \underbrace{H_1^{(3)} \geq \dots \geq H_{s_3}^{(3)}}_{\substack{\varepsilon(D_1^{(2)})\text{-regular} \\ D_1 \geq D^{(2)} \Rightarrow W_2 \leq H_{s_3}^{(3)}}} \dots \underbrace{H_1^{(t)} \geq \dots \geq H_{s_t}^{(t)}}_{\substack{\varepsilon(D_1^{(t-1)})\text{-regular} \\ D_1 \geq D^{(t-1)} \Rightarrow W_2 \leq H_{s_t}^{(t)}}}$$

Assume $D_1 < D^{(i)}$
for some $i < t-1$

$$\Rightarrow H_1^{(i)} \geq \dots \geq H_{s_i-2}^{(i)} \geq H_{s_i-1}^{(i)} \geq H_{s_i}^{(i)} \geq W_2$$

ENERGY

W₁

large energy gap

$$\Rightarrow \mathcal{E}(W_1) \lesssim \mathcal{E}(H_{s_i-2}^{(i)})$$

A Lower Bound for the Strong Arithmetic Regularity Lemma

$$\left[\begin{array}{l} W_2 \leq W_1 \\ \underbrace{D_2 \geq D_1}_{\text{codimensions}} \end{array} \right] \left\{ \begin{array}{l} f \text{ is } \mathcal{E}(D_1)\text{-regular on (almost all) cosets of } W_2; \\ \mathcal{E}(W_2) - \mathcal{E}(W_1) \leq \mathcal{E}(0) \end{array} \right. \quad \text{Strong ARL}$$

$$\mathbb{F}_2^n = \underbrace{H_0^{(1)} \geq H_1^{(1)} \geq \dots \geq H_{s_1}^{(1)}}_{\mathcal{E}(0)\text{-regular}} \underbrace{H_1^{(2)} \geq \dots \geq H_{s_2}^{(2)}}_{\mathcal{E}(D_1^{(1)})\text{-regular}} \underbrace{H_1^{(3)} \geq \dots \geq H_{s_3}^{(3)}}_{\mathcal{E}(D_1^{(2)})\text{-regular}} \dots \underbrace{H_1^{(t)} \geq \dots \geq H_{s_t}^{(t)}}_{\mathcal{E}(D_1^{(t-1)})\text{-regular}}$$

$D_1 \geq 0 \Rightarrow W_2 \leq H_{s_1}^{(1)}$
 $D_1 \geq D_1^{(1)} \Rightarrow W_2 \leq H_{s_2}^{(2)}$
 $D_1 \geq D_1^{(2)} \Rightarrow W_2 \leq H_{s_3}^{(3)}$
 $D_1 \geq D_1^{(t-1)} \Rightarrow W_2 \leq H_{s_t}^{(t)}$

Assume $D_1 < D_1^{(i)}$
for some $i < t-1$

$$\Rightarrow H_1^{(i)} \geq \dots \geq H_{s_i-2}^{(i)} \geq H_{s_i-1}^{(i)} \geq H_{s_i}^{(i)} \geq W_2$$

$$\Rightarrow \mathcal{E}(W_1) \lesssim \mathcal{E}(H_{s_i-2}^{(i)})$$

A Lower Bound for the Strong Arithmetic Regularity Lemma

Thank you for
your attention! :)

email: vg338@cam.ac.uk